Single Circle Noise Test (October 14, 2026)

Completed single-circle study · 1,440 paired cases · 4,320 reconstructions

1. Inverse problems and protocol

- **CT:** 10 equally spaced parallel-beam projection angles over $[0, \pi)$.
- **MRI:** single-coil Fourier measurements with variable-density Cartesian sampling at nominal $4\times$ acceleration.
- **Noise:** 1/5/10/20%.
- **Resolutions:** $64^2, 96^2, 128^2, 160^2, 200^2, 250^2$.
- **Samples:** 30 per resolution, inverse problem and noise level; paired across methods.
- **Models:** non-attention flow-matching UNets.
- **Euler steps:** 5.
- **Optimizer:** LBFGS.
- **Learning rate:** 1.
- **DLR rank:** $r = 8$.
- **Regularization:** none.

2. Best, median and worst relative image error

Image error (%): $100\|\hat{x} - x_{\mathrm{GT}}\|_2 / \|x_{\mathrm{GT}}\|_2$. Each row summarizes 30 final reconstructions at one resolution and noise level. Best/median/worst are computed separately for each method. Within each row, green marks the lowest best error, grey the lowest median, and red the highest worst error.

CT · 10 angles

1% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	1.86	3.97	99.87	1.69	3.43	23.65	**1.46**	**2.18**	**100.37**
96^2	2.14	3.42	115.33	1.42	3.18	14.34	**1.37**	**2.34**	**123.40**
128^2	2.79	4.18	**119.07**	1.63	3.48	15.26	**1.42**	**2.10**	117.34
160^2	1.86	4.43	100.21	**1.30**	2.56	9.55	1.31	**2.18**	**114.58**
200^2	2.54	4.76	109.66	1.20	4.65	110.54	**1.14**	**1.49**	**112.02**
250^2	1.74	5.89	99.87	1.17	2.35	**112.04**	**1.16**	**1.70**	111.06

5% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	7.16	32.77	**104.06**	2.31	4.13	40.93	**1.87**	**2.81**	101.00
96^2	6.20	28.30	116.02	**1.75**	3.64	25.75	1.80	**3.58**	**124.03**
128^2	6.34	25.00	**118.87**	1.93	4.32	32.26	**1.70**	**2.45**	116.04
160^2	2.25	27.75	101.95	**1.49**	3.60	26.42	1.66	**2.19**	**106.99**

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
200^2	5.82	29.21	108.89	2.11	9.25	110.43	**1.30**	**2.31**	**111.81**
250^2	5.59	29.19	107.22	1.61	3.27	**112.06**	**1.34**	**1.91**	111.41

10% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	5.12	57.45	101.41	2.79	6.33	76.28	**2.16**	**4.38**	**122.01**
96^2	8.54	56.04	**130.11**	2.39	4.15	53.98	**2.18**	**3.84**	124.20
128^2	6.80	51.24	116.47	2.39	4.40	59.32	**2.18**	**4.38**	**117.96**
160^2	13.01	60.62	110.32	2.15	8.45	58.73	**2.03**	**4.44**	**114.56**
200^2	8.07	69.01	**120.11**	2.74	11.88	110.50	**1.63**	**2.81**	111.84
250^2	6.26	67.33	**118.72**	1.70	3.74	112.07	**1.43**	**2.95**	111.53

20% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	8.34	120.99	**225.02**	3.04	11.59	145.06	**2.84**	**5.32**	102.50
96^2	8.48	114.15	**329.29**	3.80	9.14	133.52	**2.93**	**6.90**	124.70
128^2	9.11	111.73	**352.54**	**2.14**	8.29	108.81	3.39	**6.89**	117.70
160^2	20.55	118.42	**255.90**	3.91	12.21	114.31	**3.75**	**6.78**	114.35
200^2	12.55	157.81	**352.70**	2.75	9.75	110.79	**2.71**	**4.70**	111.80
250^2	6.91	116.39	**205.66**	2.12	37.78	112.08	**1.69**	**4.68**	111.88

MRI · $4\times$ acceleration

1% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	**1.25**	**1.46**	1.77	1.35	2.18	3.43	1.38	2.03	**100.40**
96^2	**1.07**	**1.36**	103.55	1.33	2.34	**135.32**	1.25	1.81	127.53
128^2	**1.06**	**1.16**	104.88	1.28	2.17	65.85	1.18	2.22	**114.39**
160^2	**0.93**	**1.14**	119.80	1.21	2.33	82.04	1.20	4.45	**126.95**
200^2	**0.93**	**1.06**	**106.17**	1.05	1.60	5.58	1.07	1.63	104.71
250^2	**0.90**	**1.06**	99.63	0.99	1.55	3.14	1.09	1.44	**109.67**

5% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	1.88	4.75	99.75	1.61	7.11	8.15	**1.47**	**2.04**	**100.41**
96^2	2.59	4.88	116.89	1.33	5.28	**134.72**	**1.25**	**2.08**	125.37
128^2	2.56	5.32	105.59	1.43	6.99	65.54	**1.31**	**2.64**	**109.83**
160^2	1.37	4.81	115.85	1.61	7.43	81.92	**1.24**	**2.22**	**129.26**
200^2	2.14	4.64	**105.50**	1.31	7.35	8.26	**1.18**	**2.25**	104.70
250^2	1.90	6.18	99.74	1.13	7.38	8.42	**1.09**	**1.49**	**99.97**

10% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	6.20	7.14	**100.48**	1.97	12.96	15.15	**1.46**	**2.30**	100.44
96^2	4.48	12.30	103.67	1.46	12.99	**134.70**	**1.33**	**2.02**	123.86
128^2	5.60	10.33	104.63	1.36	13.33	65.62	**1.29**	**2.16**	**109.85**
160^2	1.53	11.21	**121.53**	1.84	14.00	81.75	**1.28**	**2.37**	119.14
200^2	2.76	10.11	102.65	1.32	13.76	102.30	**1.19**	**2.24**	**104.73**
250^2	3.89	12.75	99.67	1.49	13.80	15.60	**1.13**	**1.46**	**107.92**

20% noise

Resolution	D-Flow			P-Flow			DLR D-Flow		
	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)	Best (%)	Median (%)	Worst (%)
64^2	8.05	23.34	100.35	2.09	24.33	28.66	**1.85**	**2.82**	**100.48**
96^2	4.89	23.99	103.63	1.46	24.91	**136.23**	**1.43**	**2.26**	124.53
128^2	7.18	23.09	105.00	1.65	26.58	29.26	**1.57**	**2.45**	**106.44**
160^2	1.98	23.15	113.70	2.69	26.31	81.65	**1.32**	**2.56**	**121.18**
200^2	7.48	23.82	**105.73**	1.51	25.31	29.24	**1.23**	**1.71**	104.74
250^2	5.61	25.54	99.65	1.82	25.88	29.24	**1.11**	**1.49**	**99.87**

3. Best, median and worst by DLR image error

Within each resolution, noise level and inverse problem: DLR ranks 1, 15 and 30 of 30 samples (best, lower median and worst). All methods show the same case at their final iterate.

Fixed display scales; errors use original arrays. Measurement errors reference clean $A(x_{\mathrm{GT}})$.

▸ **1% noise · best / median / worst**

▸ **5% noise · best / median / worst**

▸ **10% noise · best / median / worst**

▸ **20% noise · best / median / worst**

4. Cases with the largest DLR advantage over D-Flow

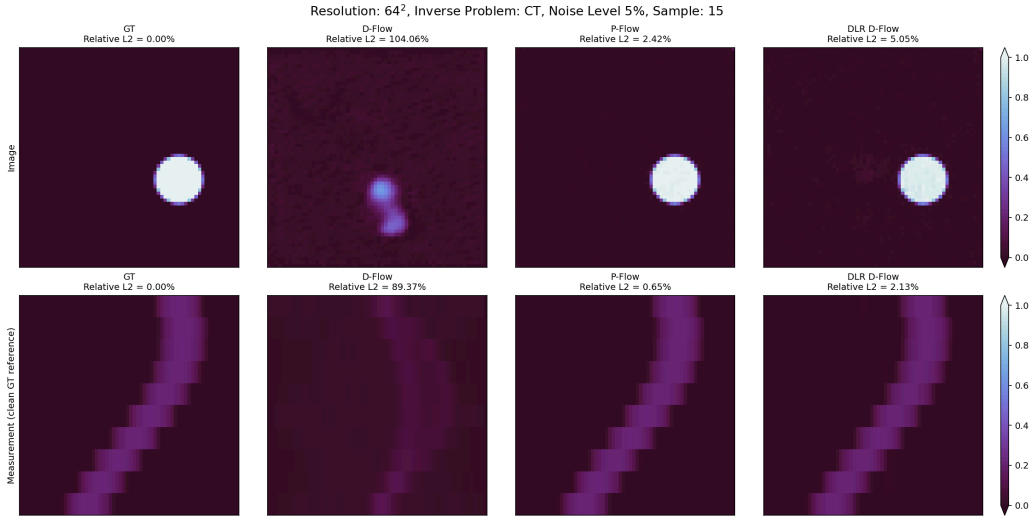
At 5%, 10% and 20% noise, show the three largest positive gaps $\Delta = e_{\text{D-Flow}} - e_{\text{DLR}}$ (percentage points), with DLR image error $\leq 10\%$, per resolution and inverse problem. $x/30$ counts cases meeting both conditions. This selection does not filter the tables above.

▼ **5% noise · DLR advantage by resolution**

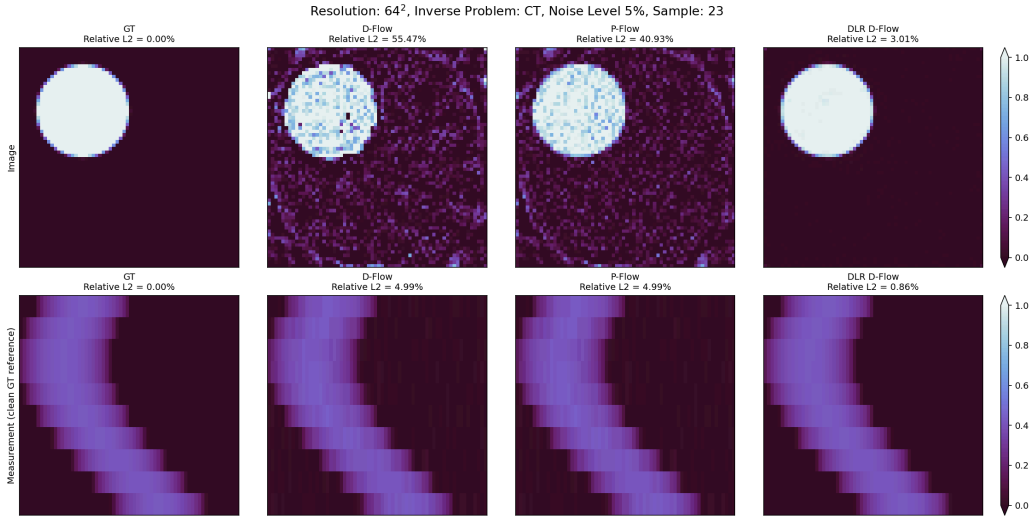
▼ **Resolution: 64^2**

▼ **CT · 27/30 qualifying cases · showing 3 · 5% noise**

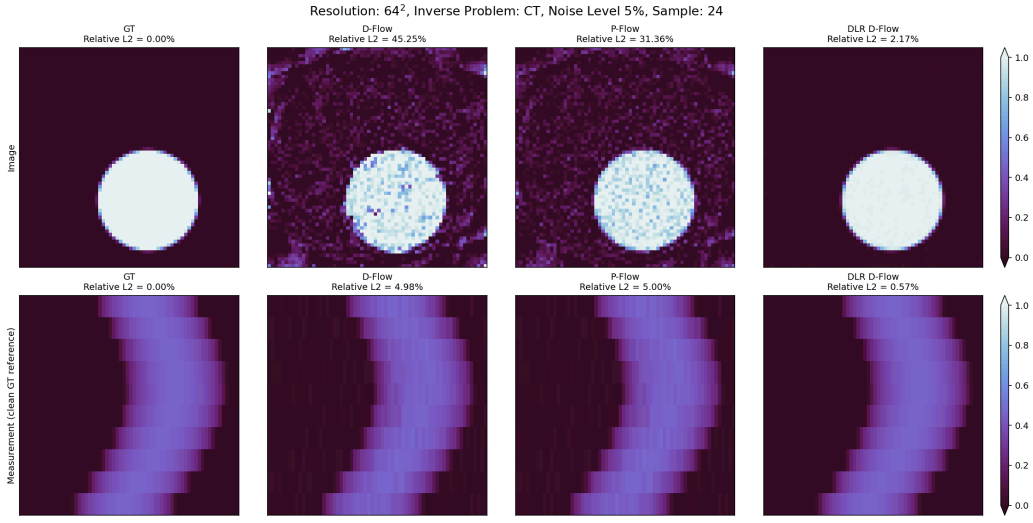
Example 1 · $\Delta = 99.01$ percentage points · CT · 64^2 · Sample 15
D-Flow: **104.06%** · P-Flow: **2.42%** · DLR D-Flow: **5.05%**



Example 2 · $\Delta = 52.47$ percentage points · CT · 64^2 · Sample 23
D-Flow: **55.47%** · P-Flow: **40.93%** · DLR D-Flow: **3.01%**



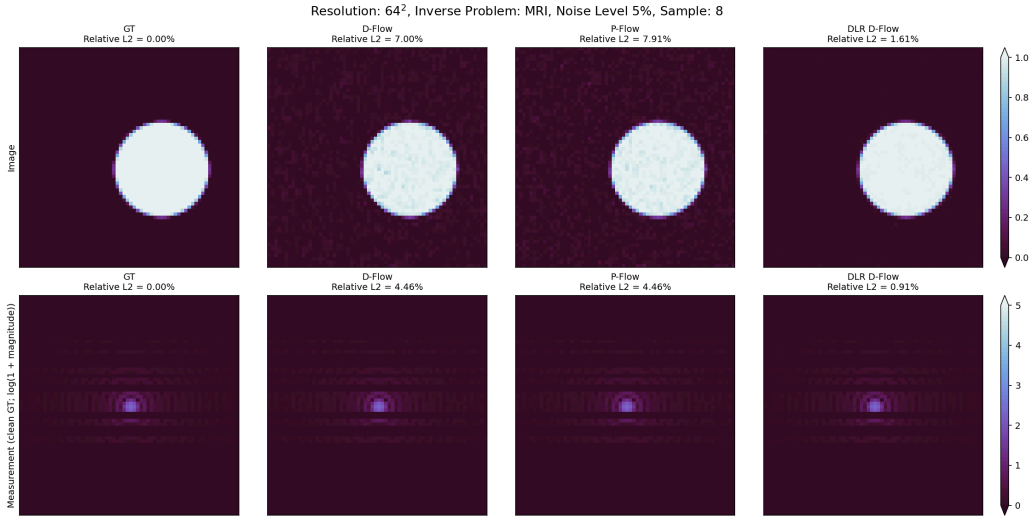
Example 3 · $\Delta = 43.08$ percentage points · CT · 64^2 · Sample 24
D-Flow: **45.25%** · P-Flow: **31.36%** · DLR D-Flow: **2.17%**



▼ MRI · 27/30 qualifying cases · showing 3 · 5% noise

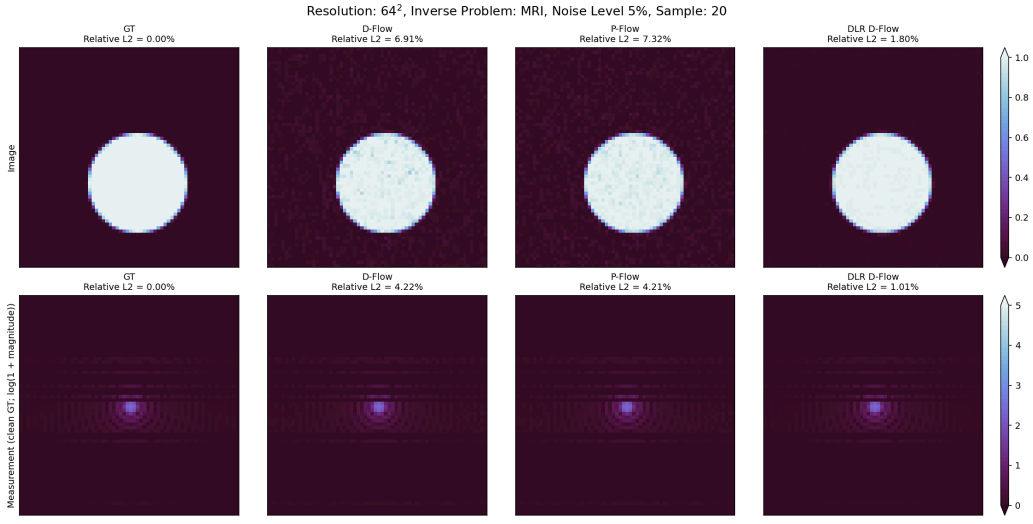
Example 1 · $\Delta = 5.39$ percentage points · MRI · 64^2 · Sample 8

D-Flow: **7.00%** · P-Flow: **7.91%** · DLR D-Flow: **1.61%**



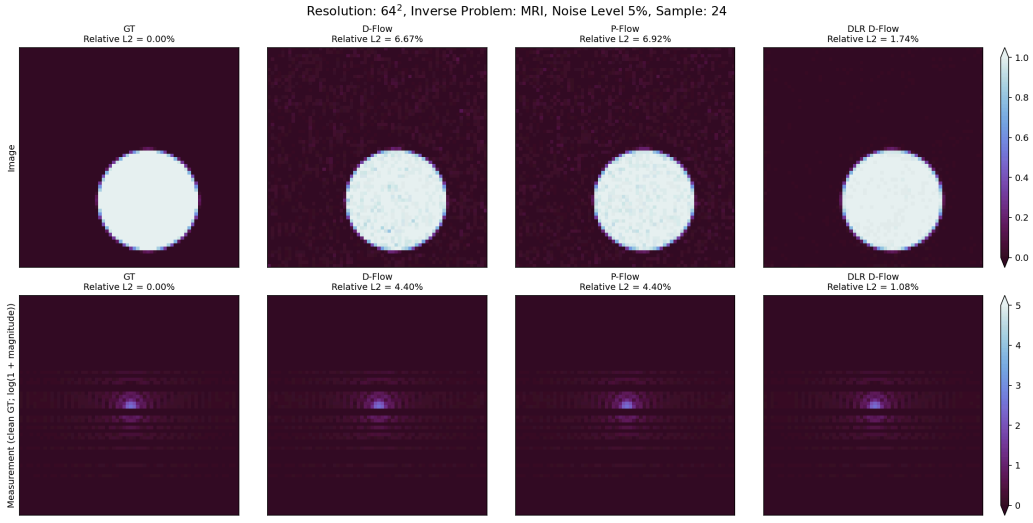
Example 2 · $\Delta = 5.11$ percentage points · MRI · 64^2 · Sample 20

D-Flow: **6.91%** · P-Flow: **7.32%** · DLR D-Flow: **1.80%**



Example 3 · $\Delta = 4.92$ percentage points · MRI · 64^2 · Sample 24

D-Flow: **6.67%** · P-Flow: **6.92%** · DLR D-Flow: **1.74%**

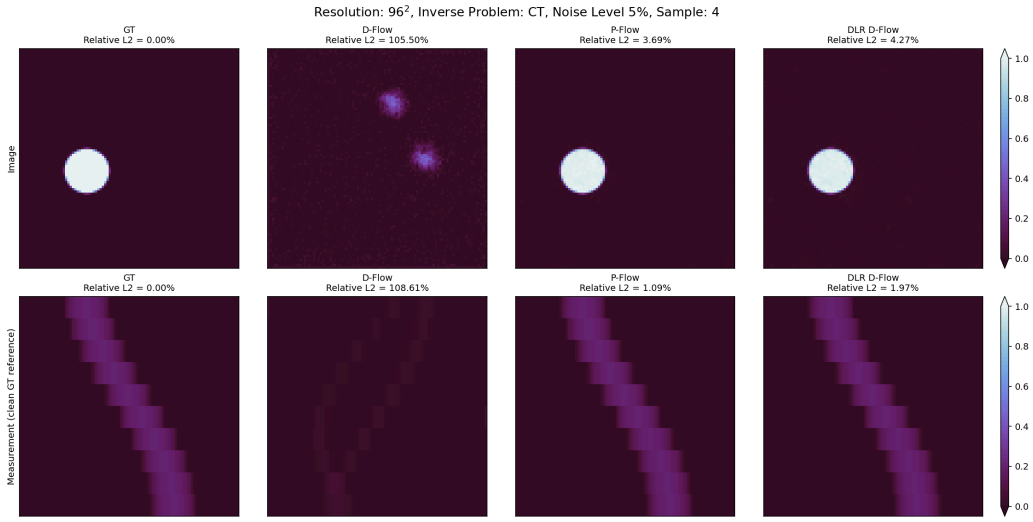


▼ Resolution: 96^2

▼ CT · 21/30 qualifying cases · showing 3 · 5% noise

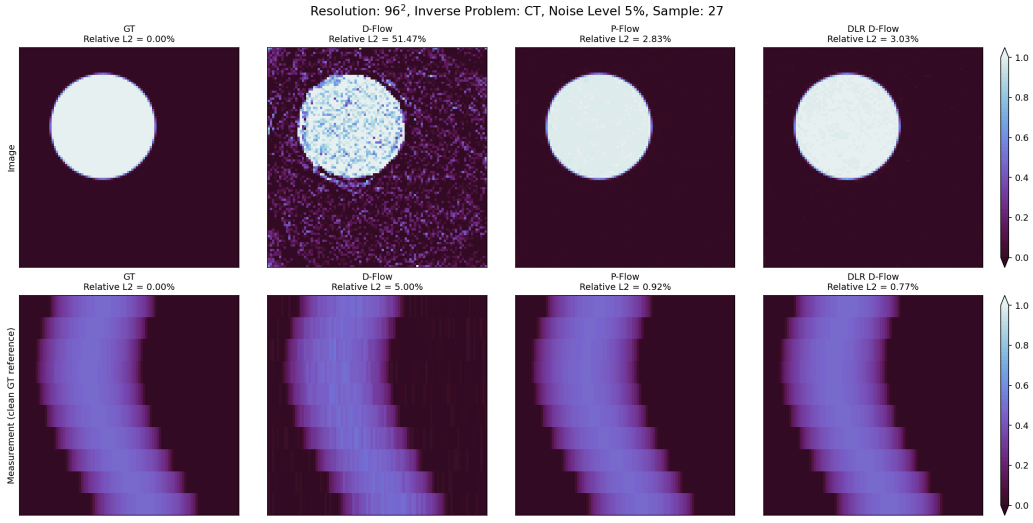
Example 1 · $\Delta = 101.23$ percentage points · CT · 96^2 · Sample 4

D-Flow: **105.50%** · P-Flow: **3.69%** · DLR D-Flow: **4.27%**



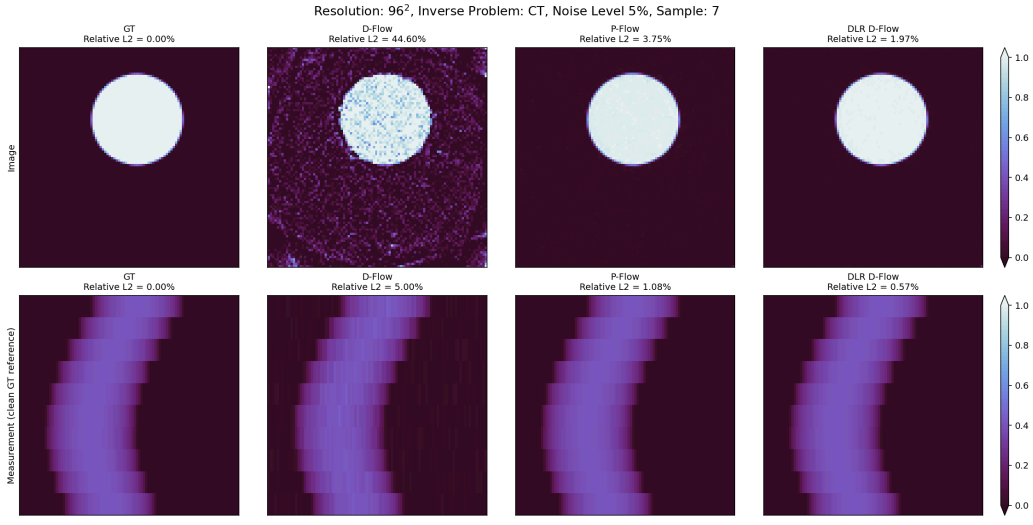
Example 2 · $\Delta = 48.44$ percentage points · CT · 96^2 · Sample 27

D-Flow: **51.47%** · P-Flow: **2.83%** · DLR D-Flow: **3.03%**



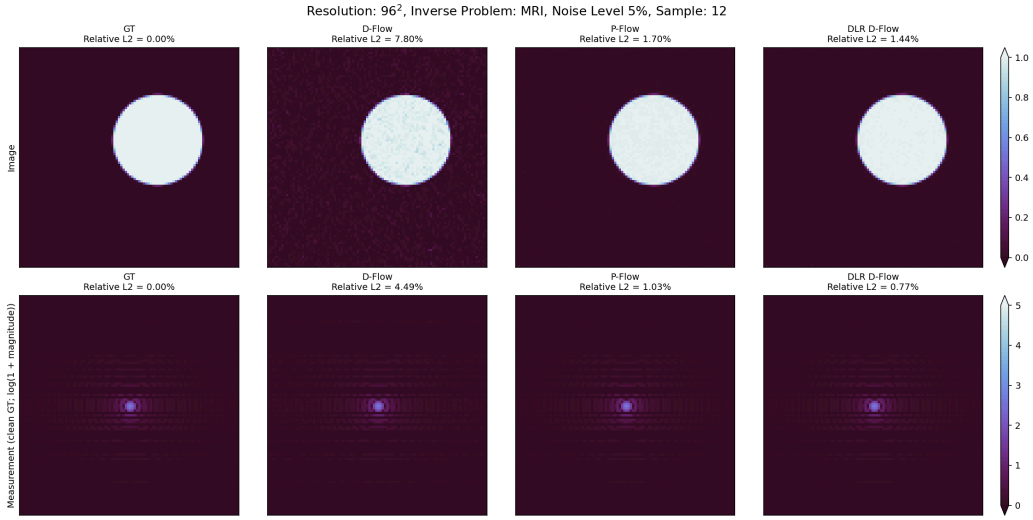
Example 3 · $\Delta = 42.63$ percentage points · CT · 96^2 · Sample 7

D-Flow: **44.60%** · P-Flow: **3.75%** · DLR D-Flow: **1.97%**

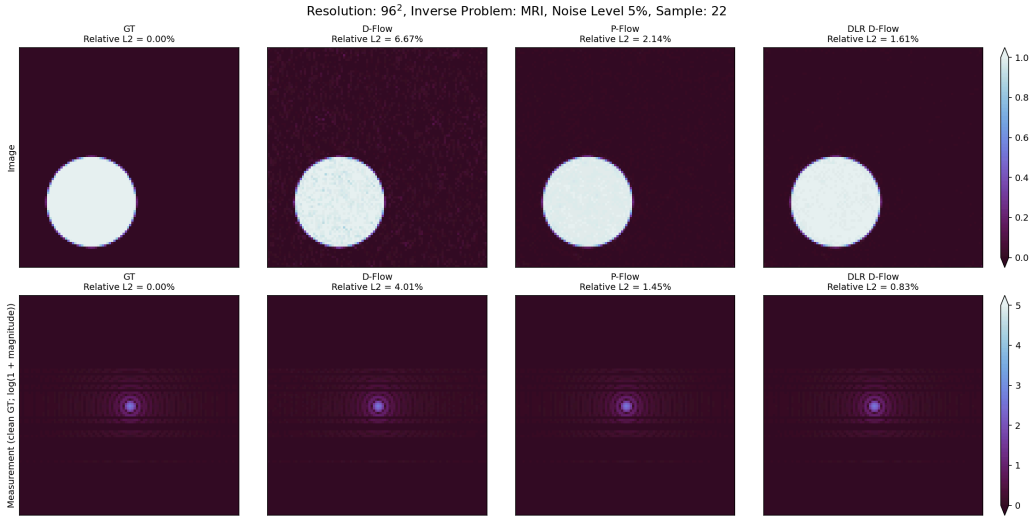


▼ MRI · 18/30 qualifying cases · showing 3 · 5% noise

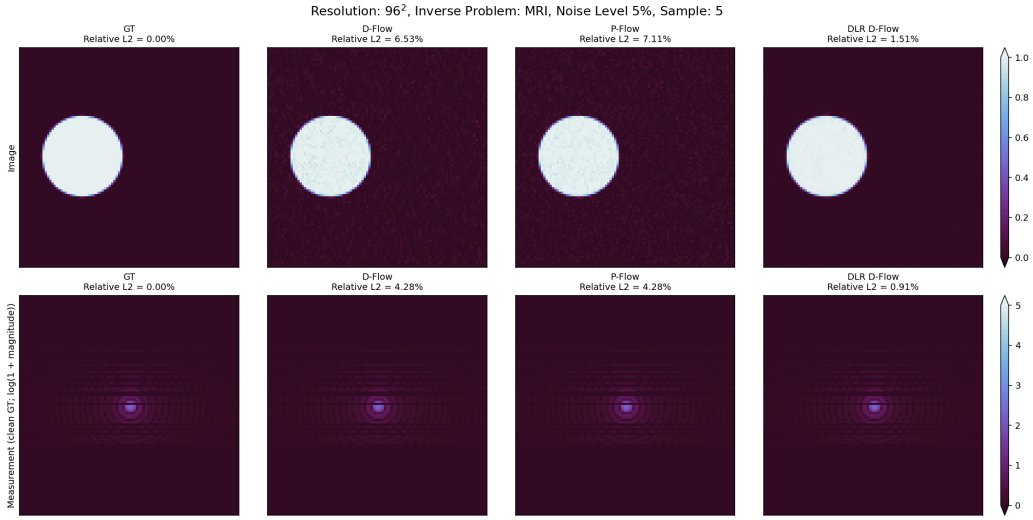
Example 1 · $\Delta = 6.36$ percentage points · MRI · 96^2 · Sample 12
D-Flow: **7.80%** · P-Flow: **1.70%** · DLR D-Flow: **1.44%**



Example 2 · $\Delta = 5.05$ percentage points · MRI · 96^2 · Sample 22
D-Flow: **6.67%** · P-Flow: **2.14%** · DLR D-Flow: **1.61%**



Example 3 · $\Delta = 5.02$ percentage points · MRI · 96^2 · Sample 5
D-Flow: **6.53%** · P-Flow: **7.11%** · DLR D-Flow: **1.51%**

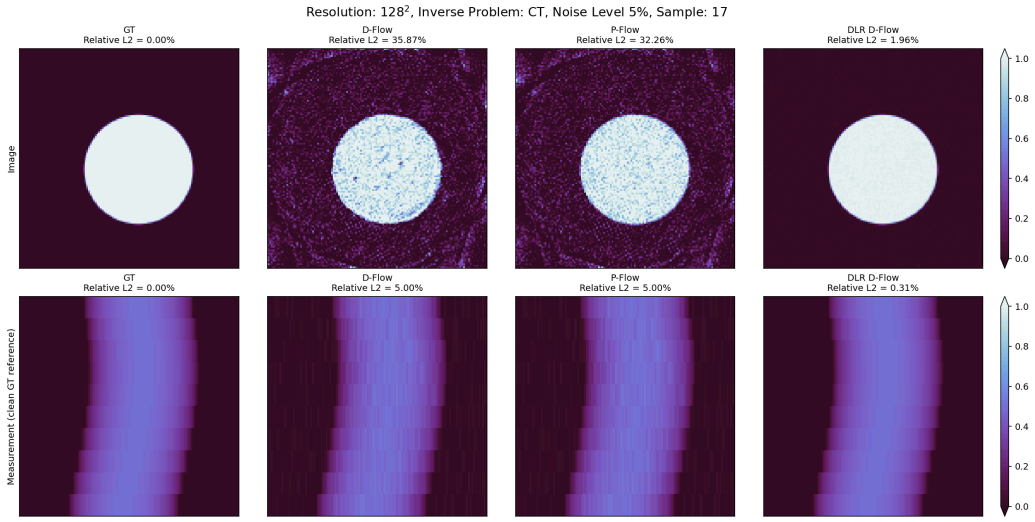


▼ Resolution: 128^2

▼ CT · 26/30 qualifying cases · showing 3 · 5% noise

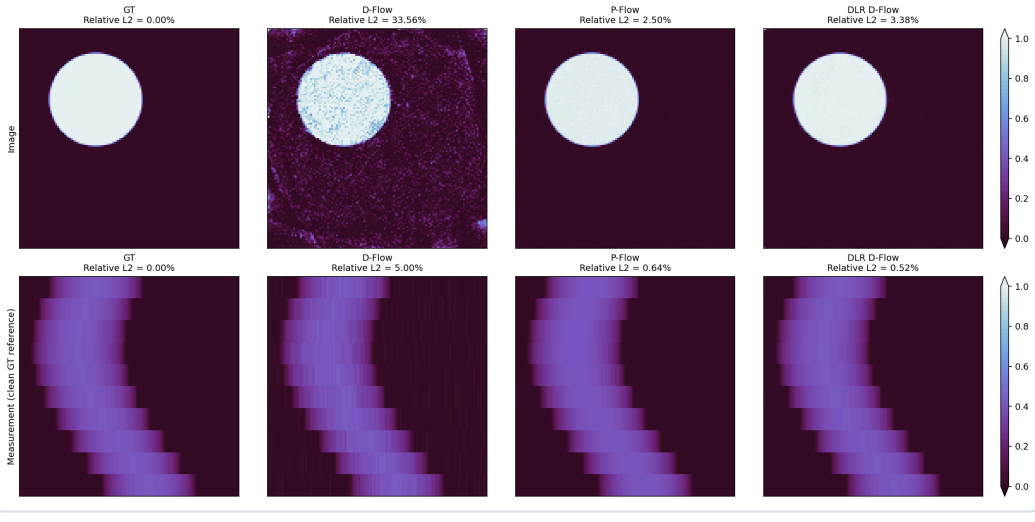
Example 1 · $\Delta = 33.91$ percentage points · CT · 128^2 · Sample 17

D-Flow: **35.87%** · P-Flow: **32.26%** · DLR D-Flow: **1.96%**



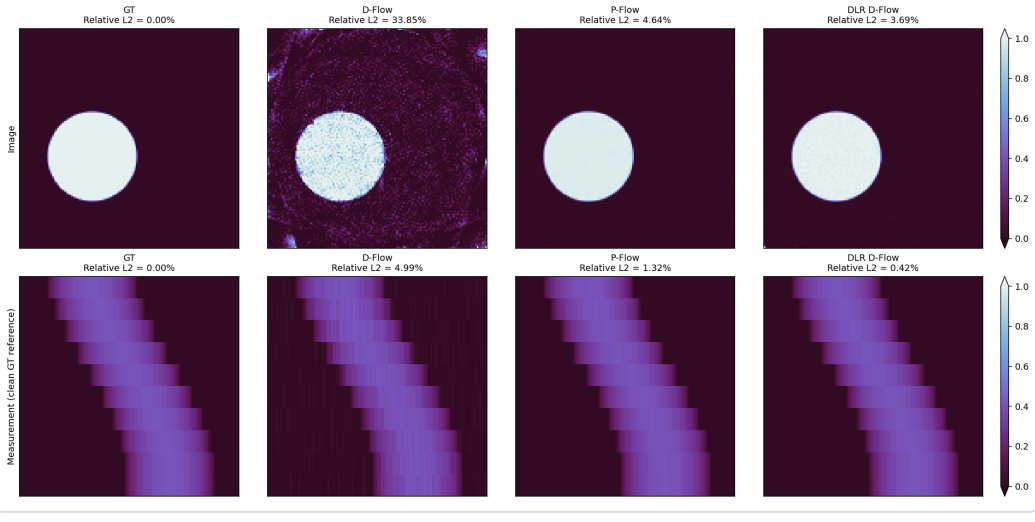
Example 2 · $\Delta = 30.17$ percentage points · CT · 128^2 · Sample 28

D-Flow: **33.56%** · P-Flow: **2.50%** · DLR D-Flow: **3.38%**



Example 3 · $\Delta = 30.16$ percentage points · CT · 128^2 · Sample 7

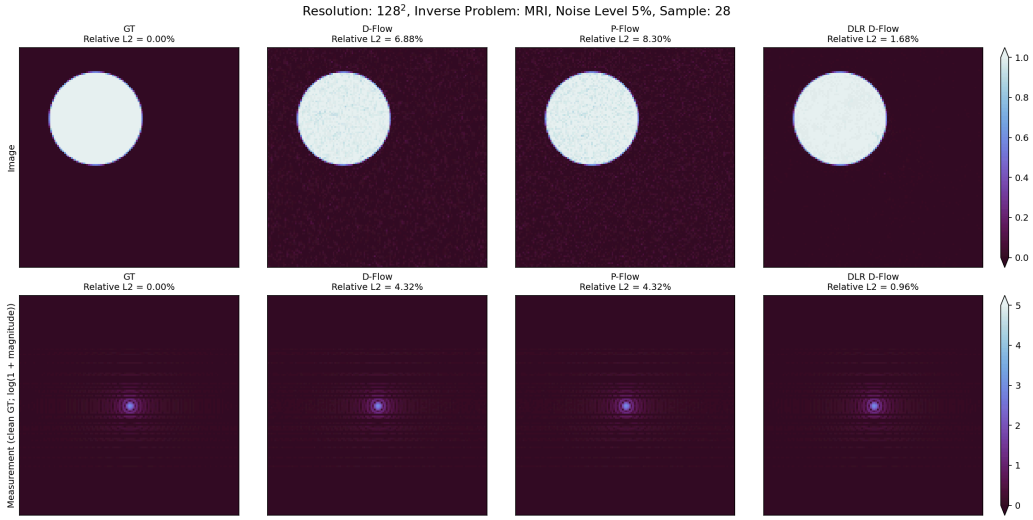
D-Flow: **33.85%** · P-Flow: **4.64%** · DLR D-Flow: **3.69%**



▼ MRI · 20/30 qualifying cases · showing 3 · 5% noise

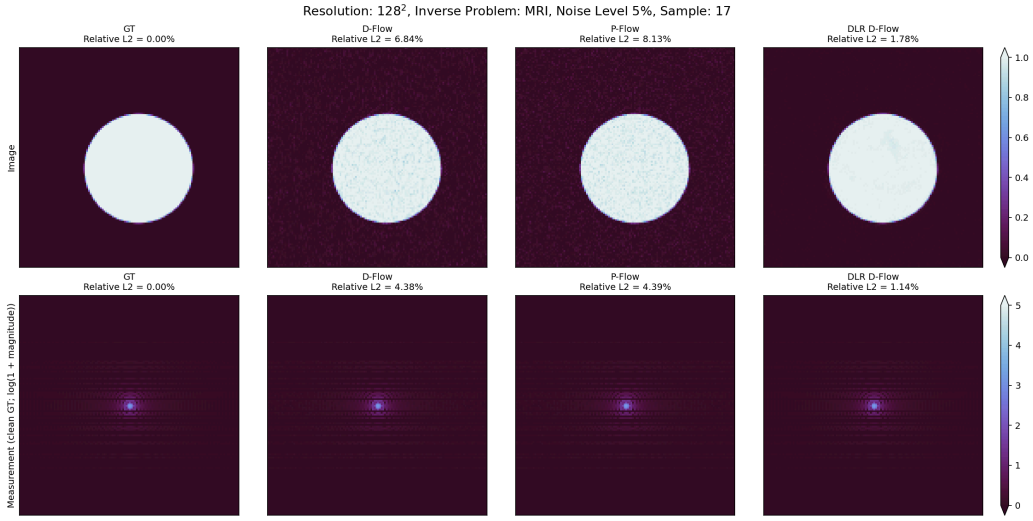
Example 1 · $\Delta = 5.20$ percentage points · MRI · 128^2 · Sample 28

D-Flow: **6.88%** · P-Flow: **8.30%** · DLR D-Flow: **1.68%**



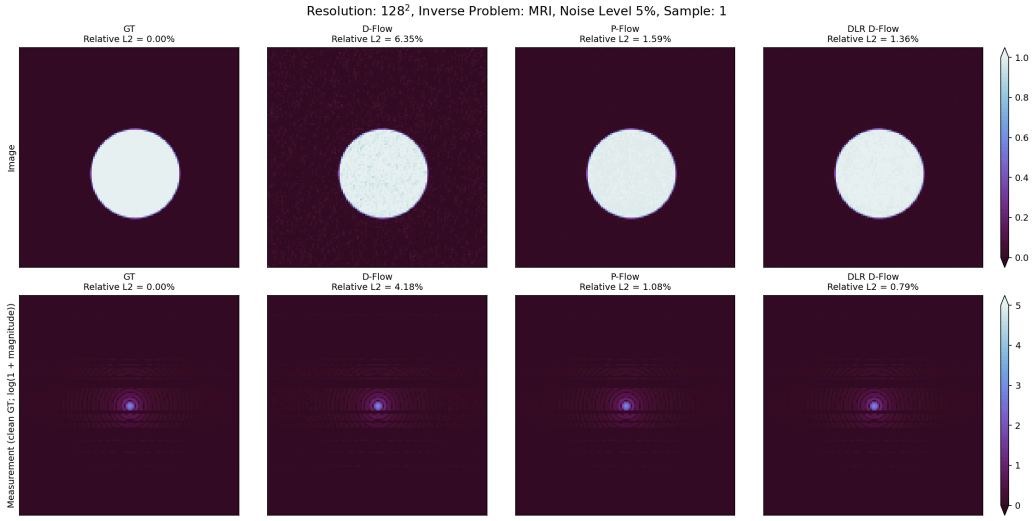
Example 2 · $\Delta = 5.05$ percentage points · MRI · 128^2 · Sample 17

D-Flow: **6.84%** · P-Flow: **8.13%** · DLR D-Flow: **1.78%**



Example 3 · $\Delta = 4.99$ percentage points · MRI · 128^2 · Sample 1

D-Flow: **6.35%** · P-Flow: **1.59%** · DLR D-Flow: **1.36%**

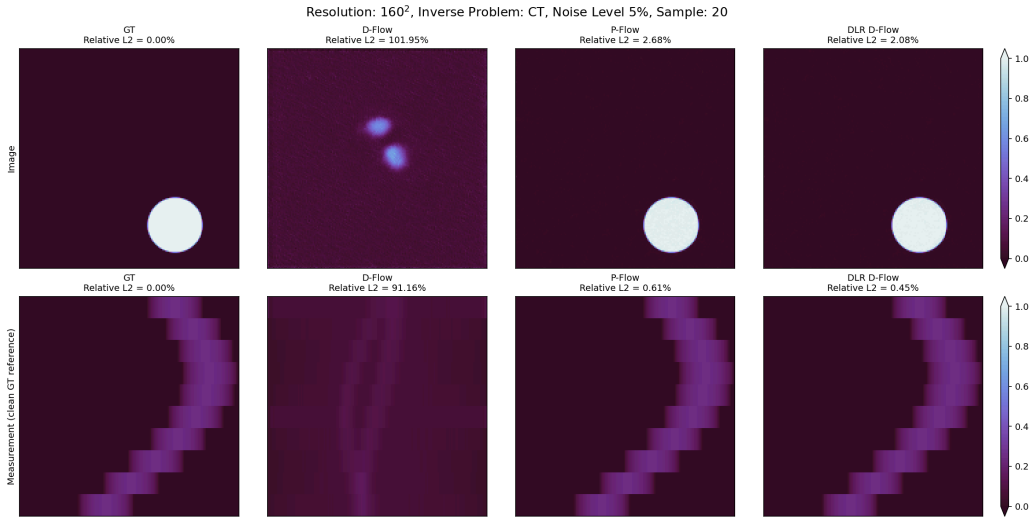


▼ Resolution: 160^2

▼ CT · 26/30 qualifying cases · showing 3 · 5% noise

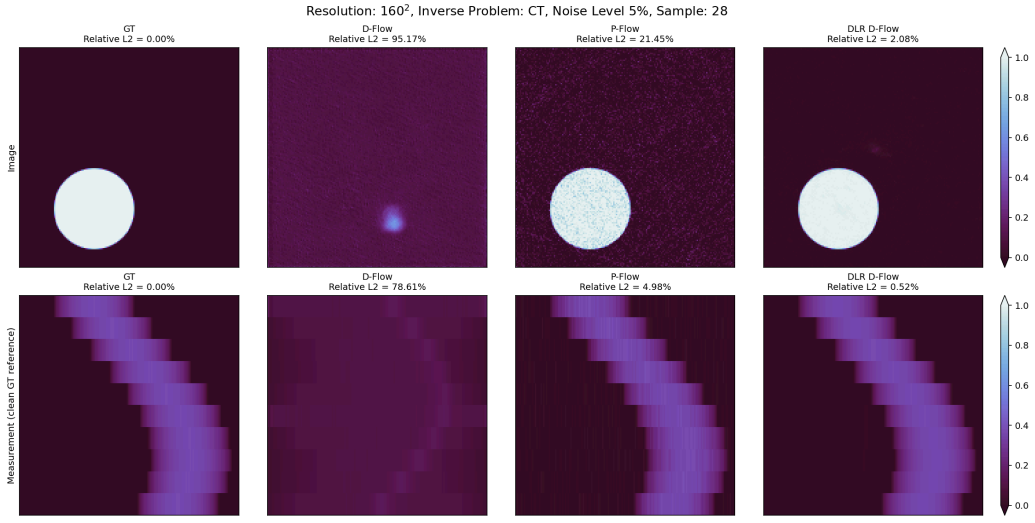
Example 1 · $\Delta = 99.87$ percentage points · CT · 160^2 · Sample 20

D-Flow: **101.95%** · P-Flow: **2.68%** · DLR D-Flow: **2.08%**



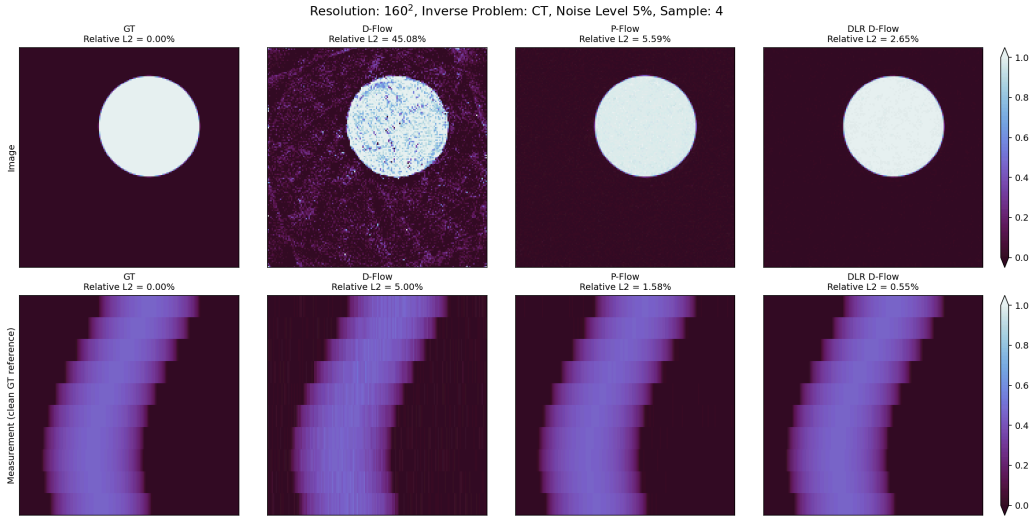
Example 2 · $\Delta = 93.09$ percentage points · CT · 160^2 · Sample 28

D-Flow: **95.17%** · P-Flow: **21.45%** · DLR D-Flow: **2.08%**



Example 3 · $\Delta = 42.43$ percentage points · CT · 160^2 · Sample 4

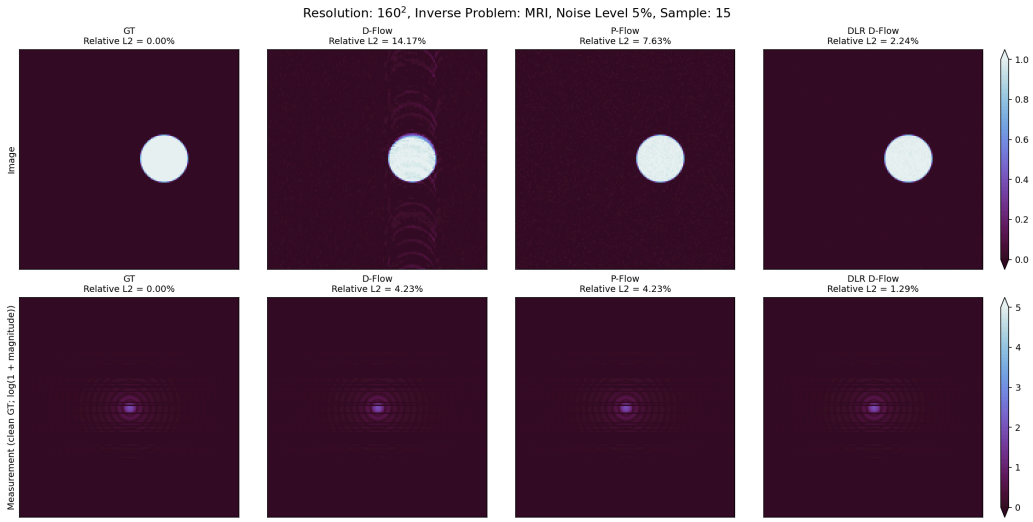
D-Flow: **45.08%** · P-Flow: **5.59%** · DLR D-Flow: **2.65%**



▼ MRI · 17/30 qualifying cases · showing 3 · 5% noise

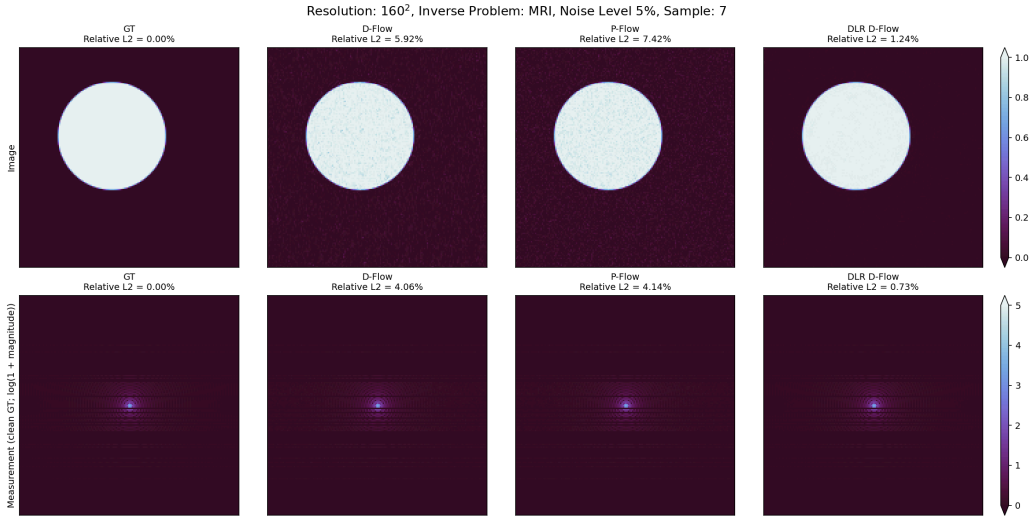
Example 1 · $\Delta = 11.93$ percentage points · MRI · 160^2 · Sample 15

D-Flow: **14.17%** · P-Flow: **7.63%** · DLR D-Flow: **2.24%**



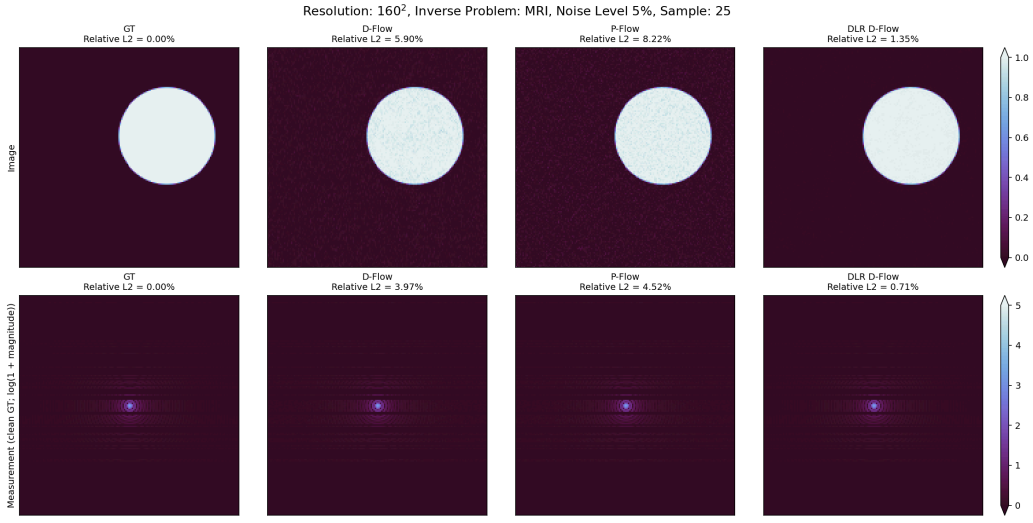
Example 2 · $\Delta = 4.68$ percentage points · MRI · 160^2 · Sample 7

D-Flow: **5.92%** · P-Flow: **7.42%** · DLR D-Flow: **1.24%**



Example 3 · $\Delta = 4.55$ percentage points · MRI · 160^2 · Sample 25

D-Flow: **5.90%** · P-Flow: **8.22%** · DLR D-Flow: **1.35%**

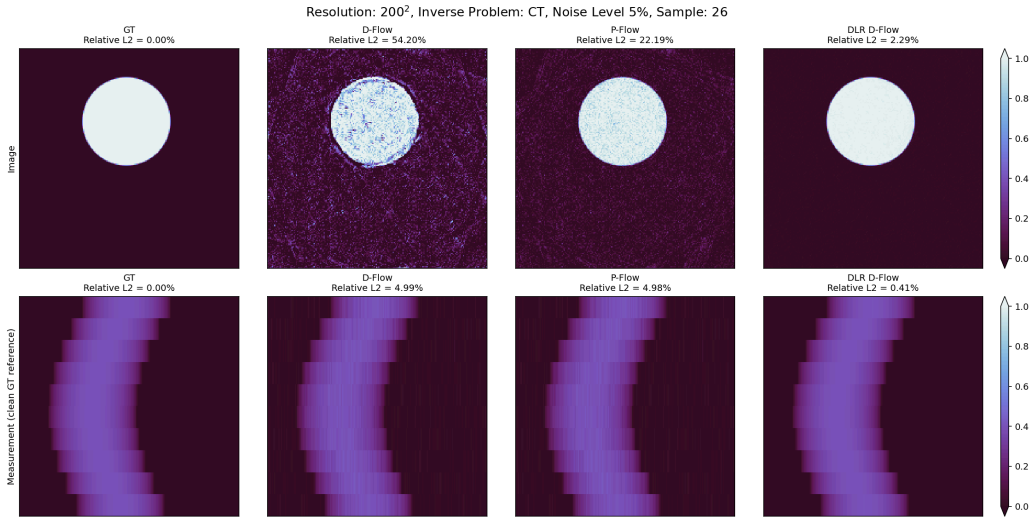


▼ Resolution: 200^2

▼ CT · 24/30 qualifying cases · showing 3 · 5% noise

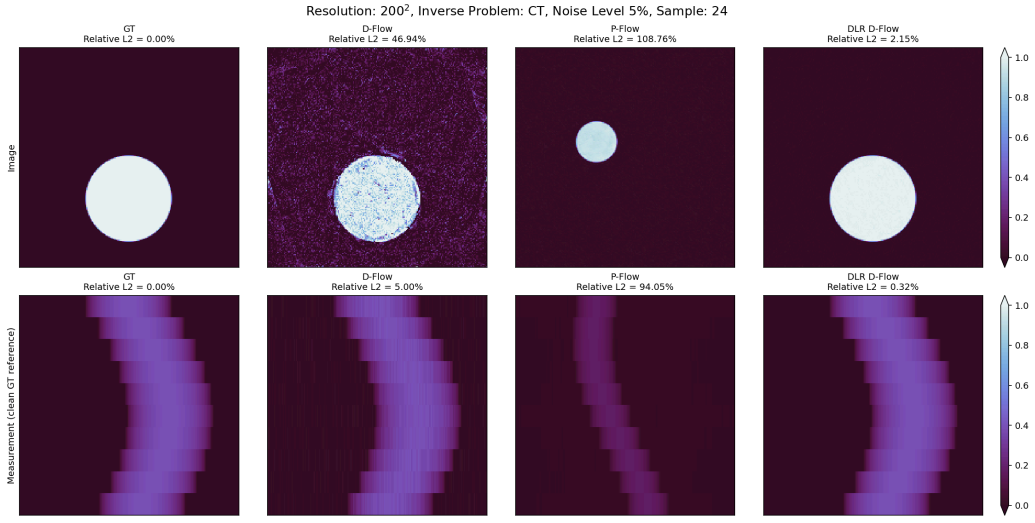
Example 1 · $\Delta = 51.91$ percentage points · CT · 200^2 · Sample 26

D-Flow: **54.20%** · P-Flow: **22.19%** · DLR D-Flow: **2.29%**



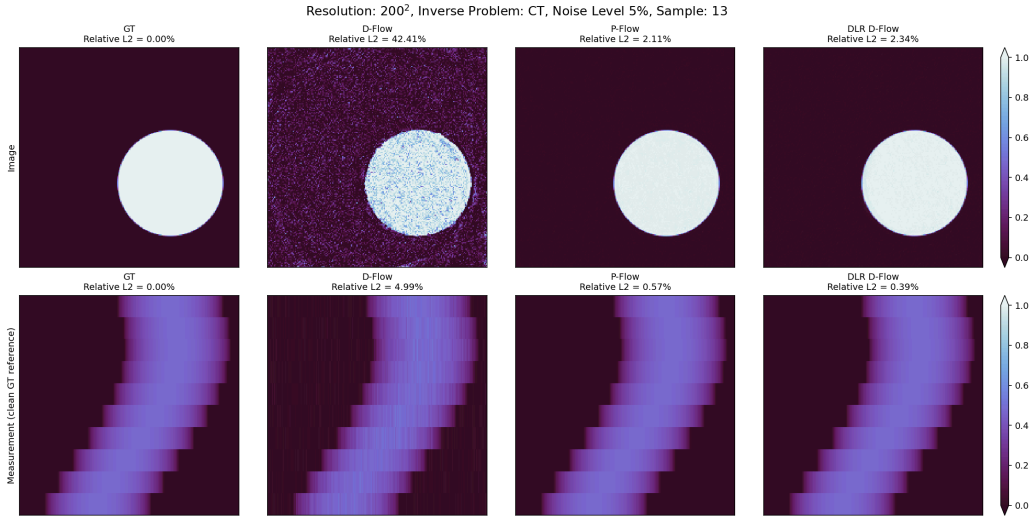
Example 2 · $\Delta = 44.79$ percentage points · CT · 200^2 · Sample 24

D-Flow: **46.94%** · P-Flow: **108.76%** · DLR D-Flow: **2.15%**



Example 3 · $\Delta = 40.08$ percentage points · CT · 200^2 · Sample 13

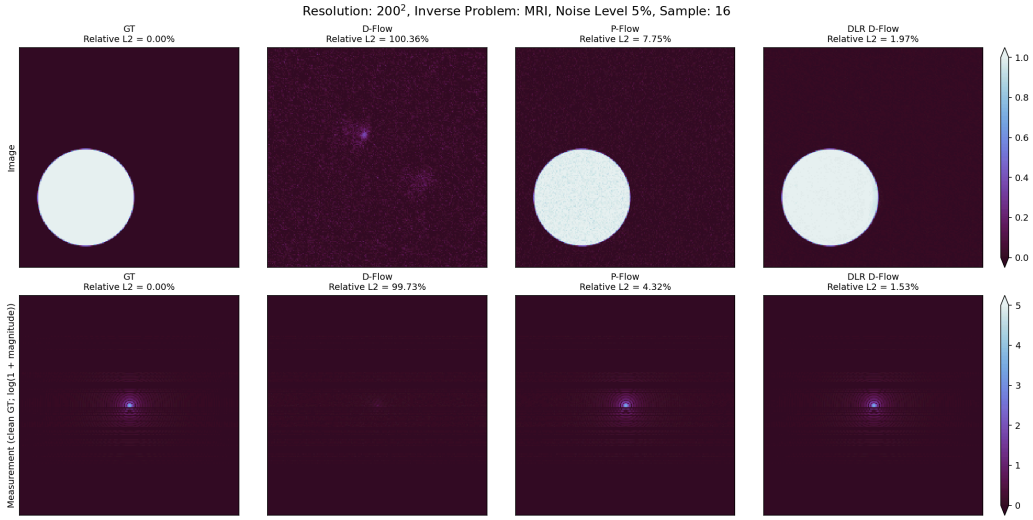
D-Flow: **42.41%** · P-Flow: **2.11%** · DLR D-Flow: **2.34%**



▼ MRI · 18/30 qualifying cases · showing 3 · 5% noise

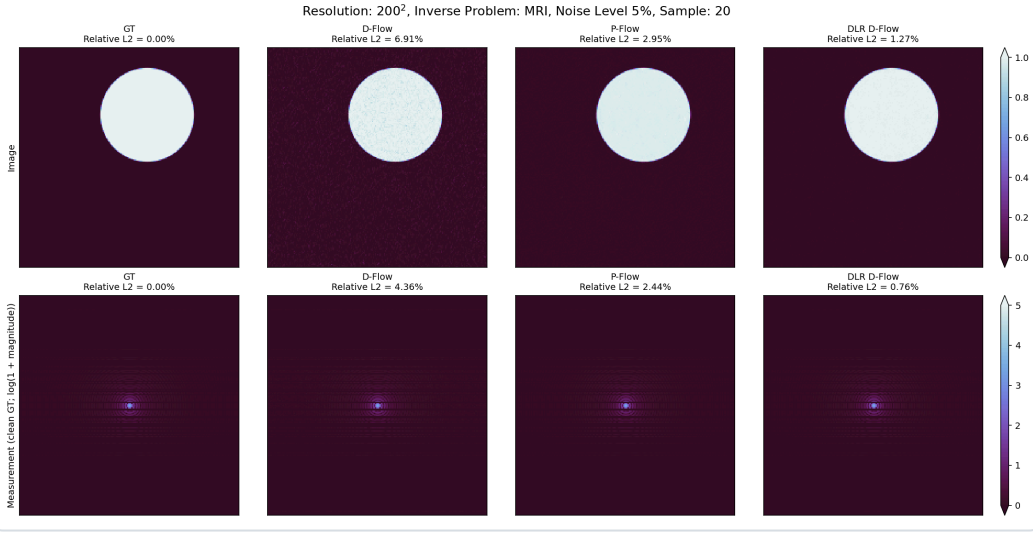
Example 1 · $\Delta = 98.40$ percentage points · MRI · 200^2 · Sample 16

D-Flow: **100.36%** · P-Flow: **7.75%** · DLR D-Flow: **1.97%**



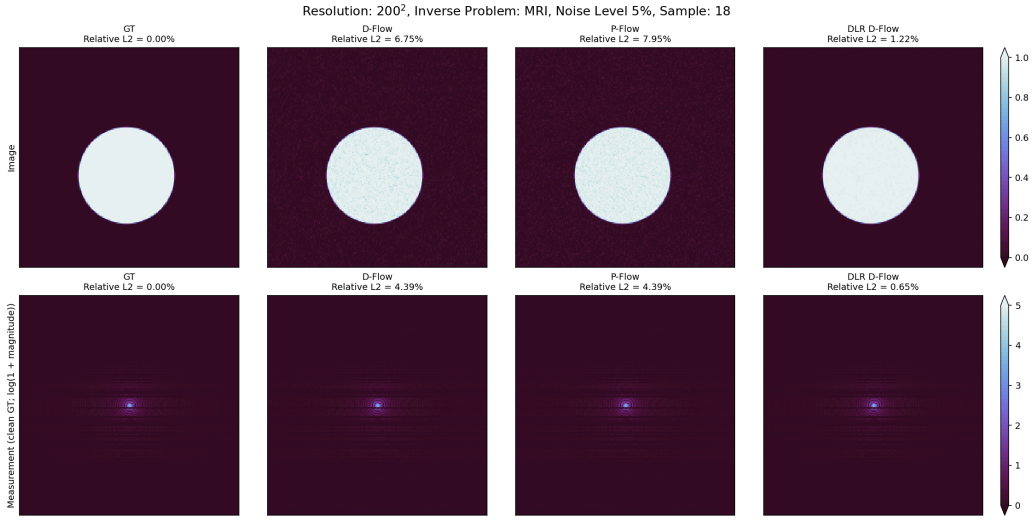
Example 2 · $\Delta = 5.64$ percentage points · MRI · 200^2 · Sample 20

D-Flow: **6.91%** · P-Flow: **2.95%** · DLR D-Flow: **1.27%**



Example 3 · $\Delta = 5.53$ percentage points · MRI · 200^2 · Sample 18

D-Flow: **6.75%** · P-Flow: **7.95%** · DLR D-Flow: **1.22%**

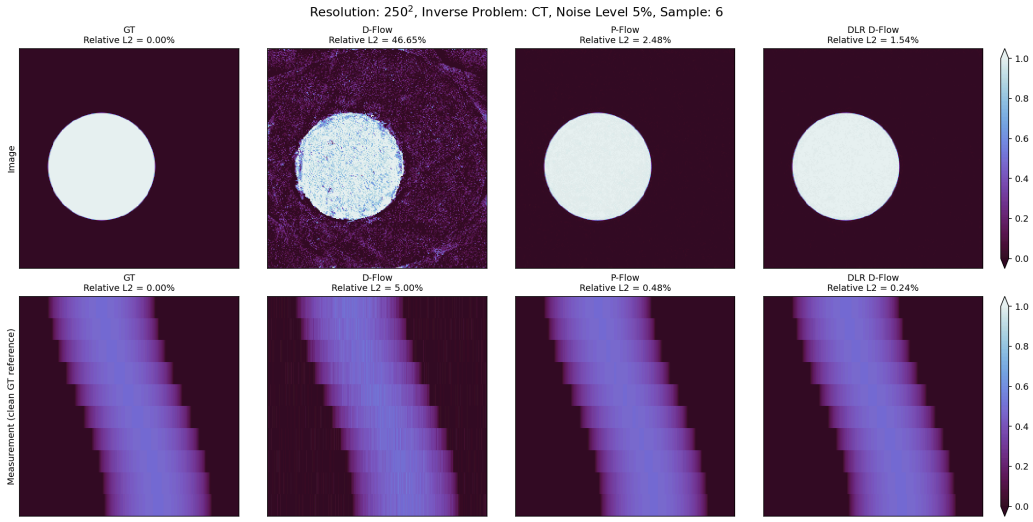


▼ Resolution: 250^2

▼ CT · 25/30 qualifying cases · showing 3 · 5% noise

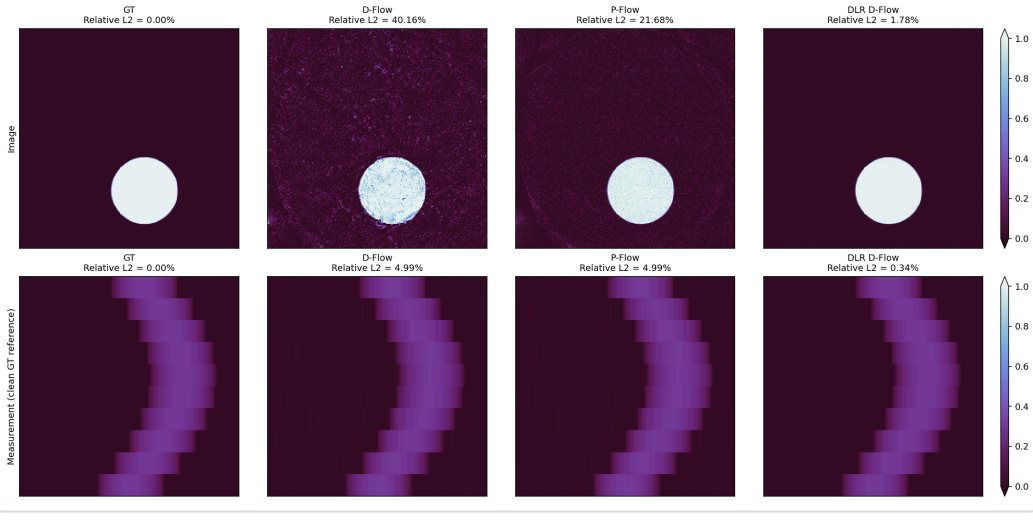
Example 1 · $\Delta = 45.11$ percentage points · CT · 250^2 · Sample 6

D-Flow: **46.65%** · P-Flow: **2.48%** · DLR D-Flow: **1.54%**



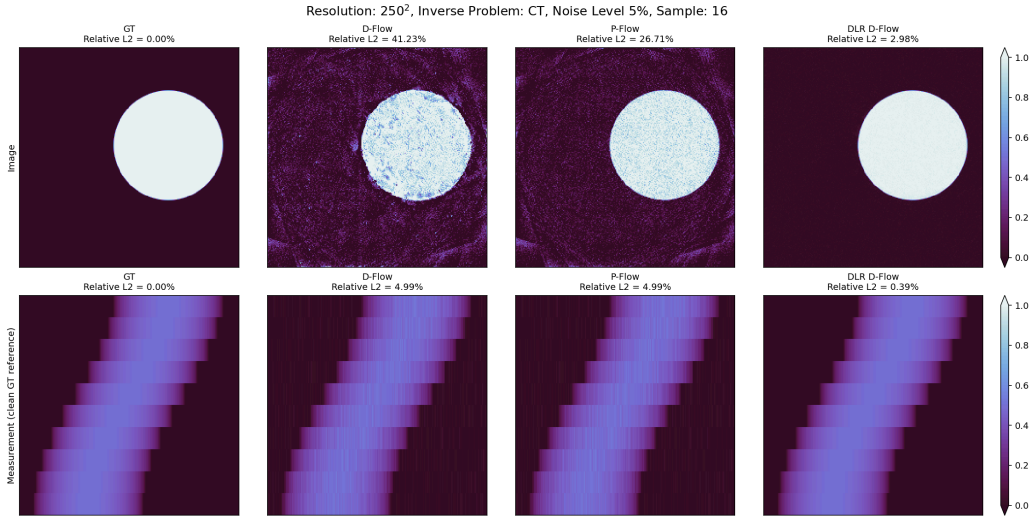
Example 2 · $\Delta = 38.38$ percentage points · CT · 250^2 · Sample 8

D-Flow: **40.16%** · P-Flow: **21.68%** · DLR D-Flow: **1.78%**



Example 3 · $\Delta = 38.25$ percentage points · CT · 250^2 · Sample 16

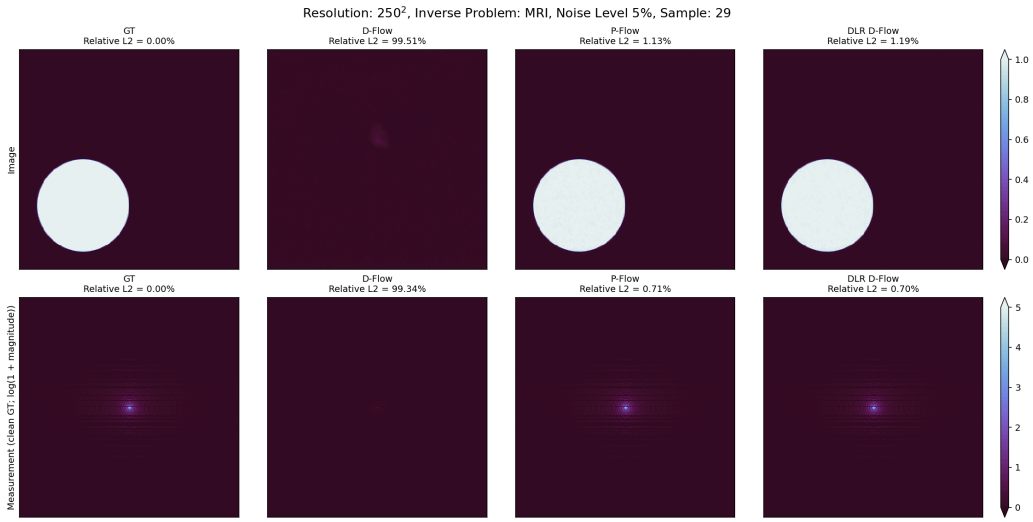
D-Flow: **41.23%** · P-Flow: **26.71%** · DLR D-Flow: **2.98%**



▼ MRI · 20/30 qualifying cases · showing 3 · 5% noise

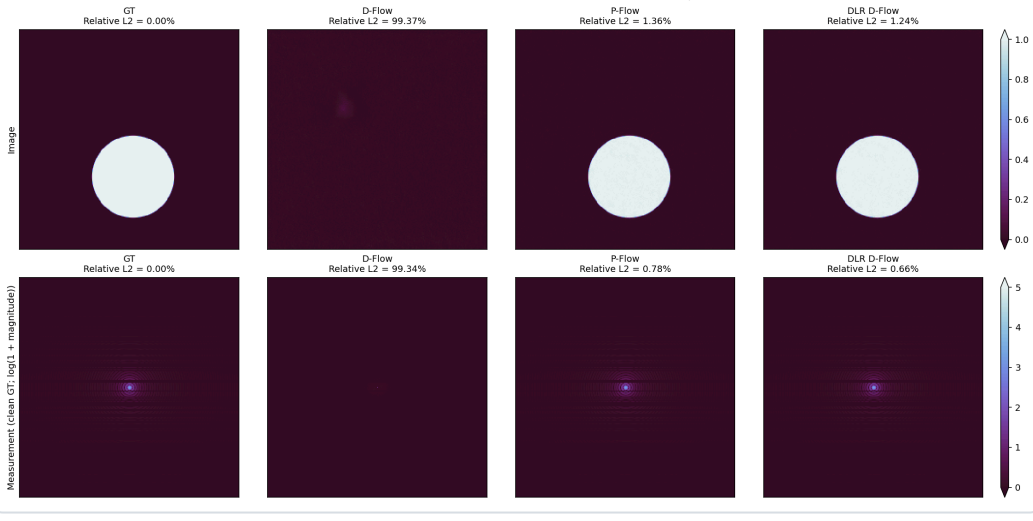
Example 1 · $\Delta = 98.32$ percentage points · MRI · 250^2 · Sample 29

D-Flow: **99.51%** · P-Flow: **1.13%** · DLR D-Flow: **1.19%**



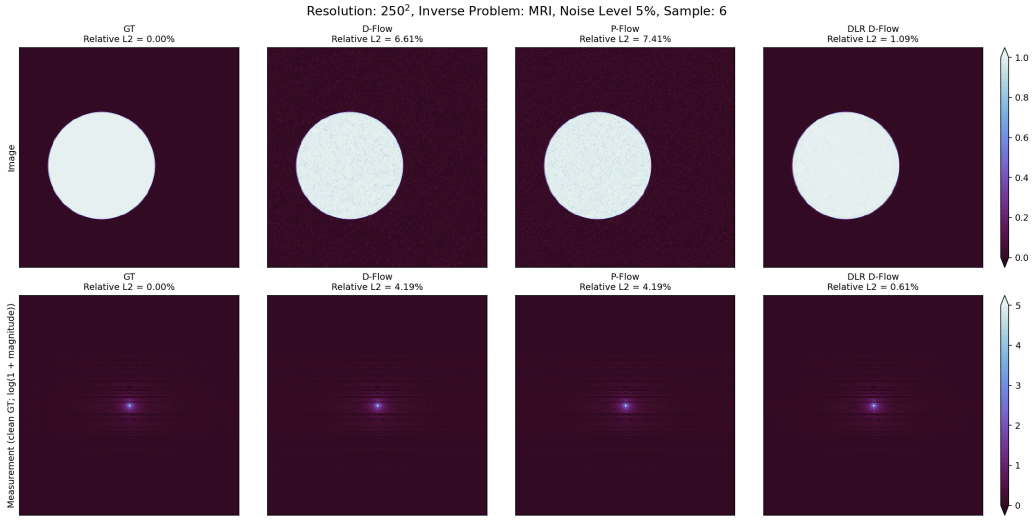
Example 2 · $\Delta = 98.13$ percentage points · MRI · 250^2 · Sample 20

D-Flow: **99.37%** · P-Flow: **1.36%** · DLR D-Flow: **1.24%**



Example 3 · $\Delta = 5.52$ percentage points · MRI · 250^2 · Sample 6

D-Flow: **6.61%** · P-Flow: **7.41%** · DLR D-Flow: **1.09%**

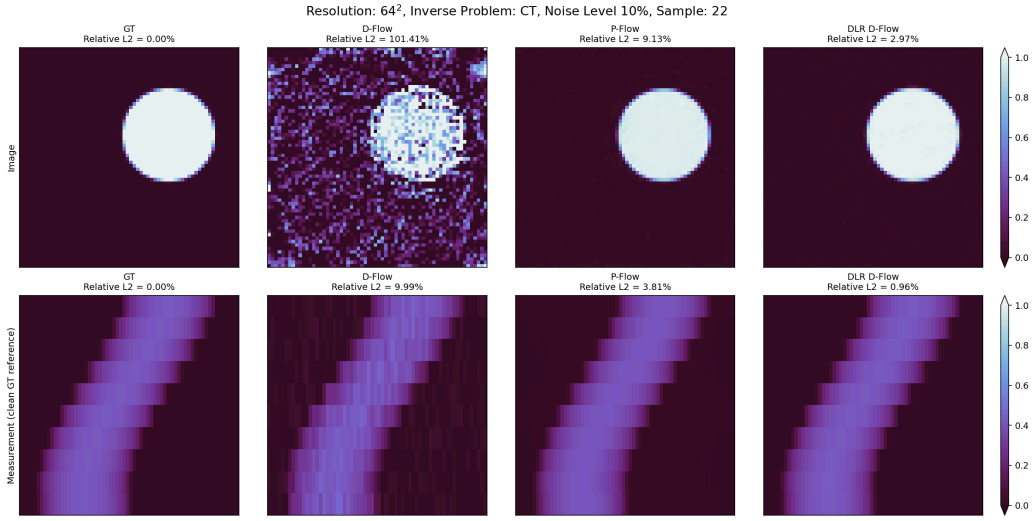


▼ **10% noise · DLR advantage by resolution**

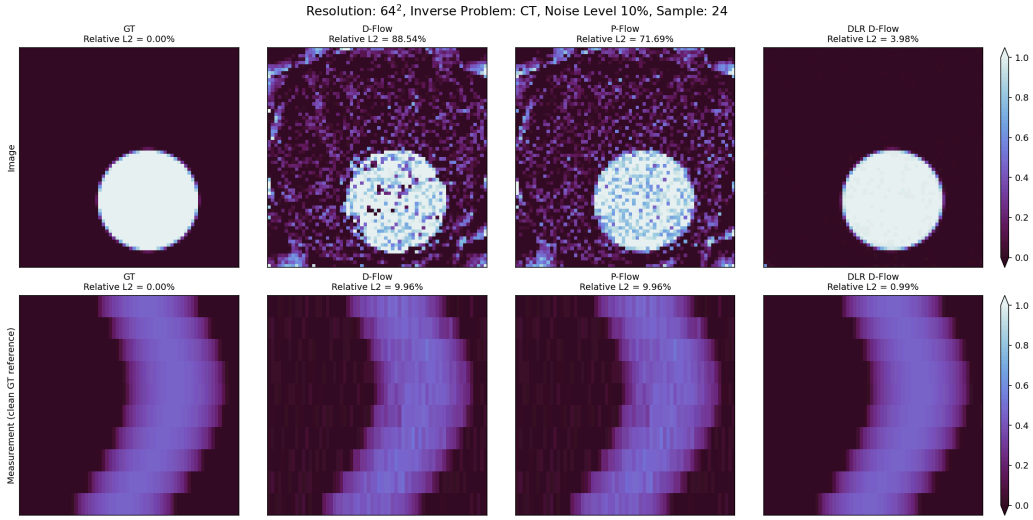
▼ **Resolution: 64^2**

▼ **CT · 27/30 qualifying cases · showing 3 · 10% noise**

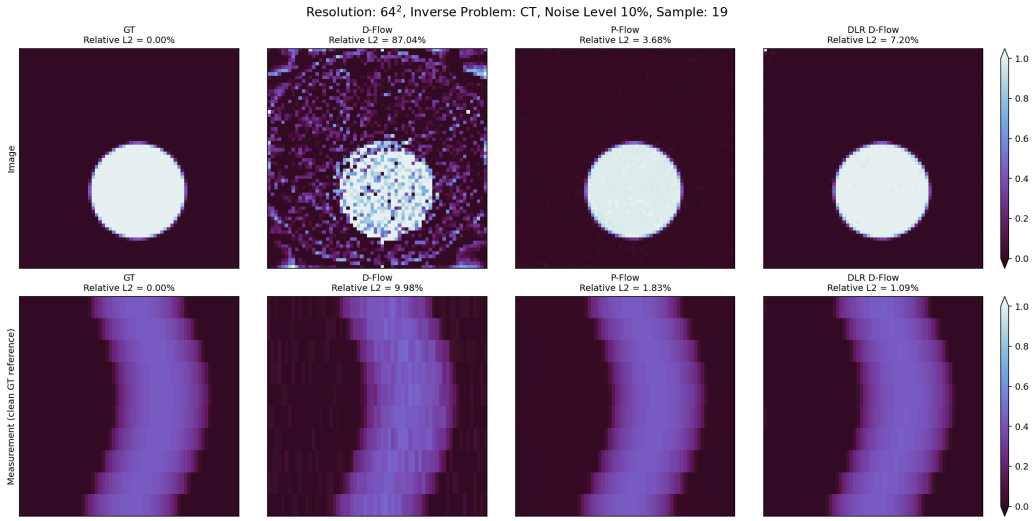
Example 1 · $\Delta = 98.44$ percentage points · CT · 64^2 · Sample 22
D-Flow: **101.41%** · P-Flow: **9.13%** · DLR D-Flow: **2.97%**



Example 2 · $\Delta = 84.55$ percentage points · CT · 64^2 · Sample 24
D-Flow: **88.54%** · P-Flow: **71.69%** · DLR D-Flow: **3.98%**



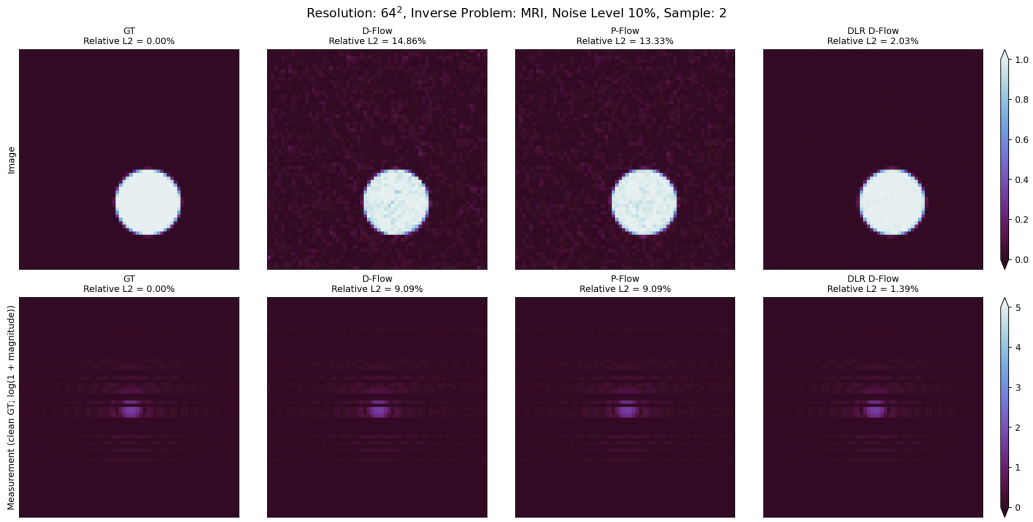
Example 3 · $\Delta = 79.84$ percentage points · CT · 64^2 · Sample 19
D-Flow: **87.04%** · P-Flow: **3.68%** · DLR D-Flow: **7.20%**



▼ MRI · 25/30 qualifying cases · showing 3 · 10% noise

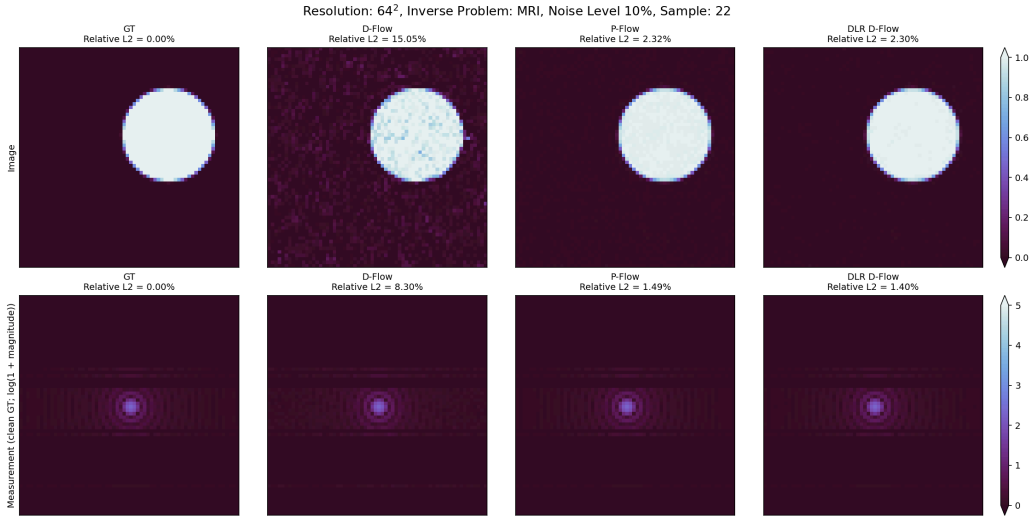
Example 1 · $\Delta = 12.83$ percentage points · MRI · 64^2 · Sample 2

D-Flow: **14.86%** · P-Flow: **13.33%** · DLR D-Flow: **2.03%**



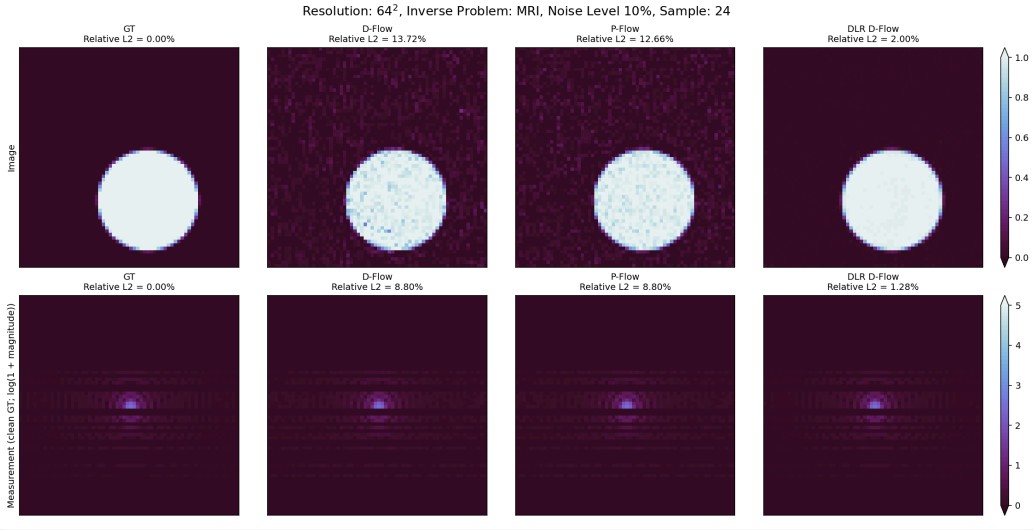
Example 2 · $\Delta = 12.76$ percentage points · MRI · 64^2 · Sample 22

D-Flow: **15.05%** · P-Flow: **2.32%** · DLR D-Flow: **2.30%**



Example 3 · $\Delta = 11.72$ percentage points · MRI · 64^2 · Sample 24

D-Flow: **13.72%** · P-Flow: **12.66%** · DLR D-Flow: **2.00%**

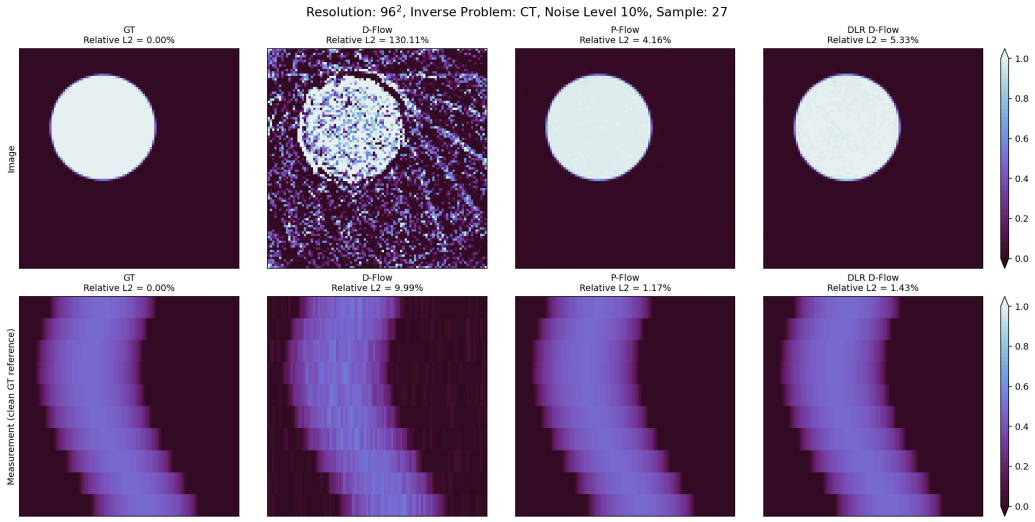


▼ Resolution: 96^2

▼ CT · 23/30 qualifying cases · showing 3 · 10% noise

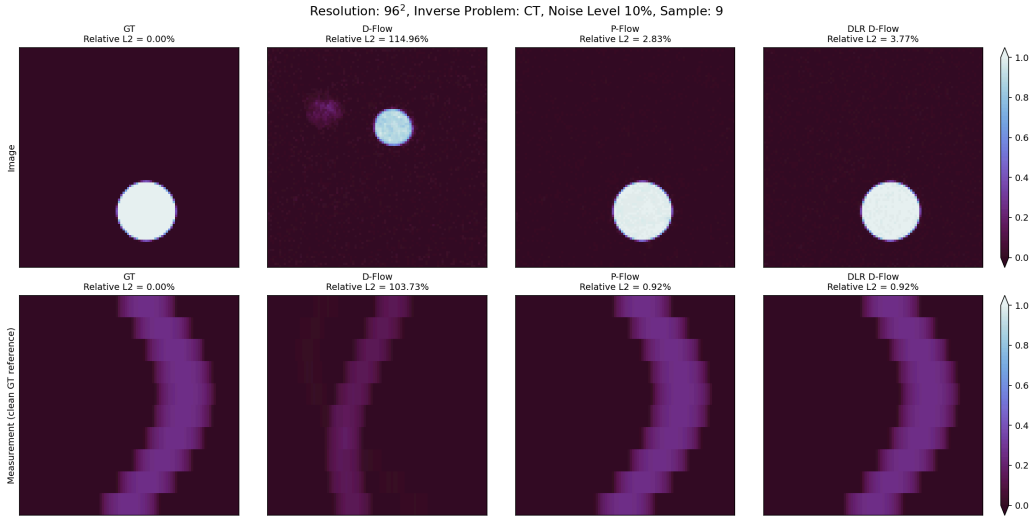
Example 1 · $\Delta = 124.78$ percentage points · CT · 96^2 · Sample 27

D-Flow: **130.11%** · P-Flow: **4.16%** · DLR D-Flow: **5.33%**



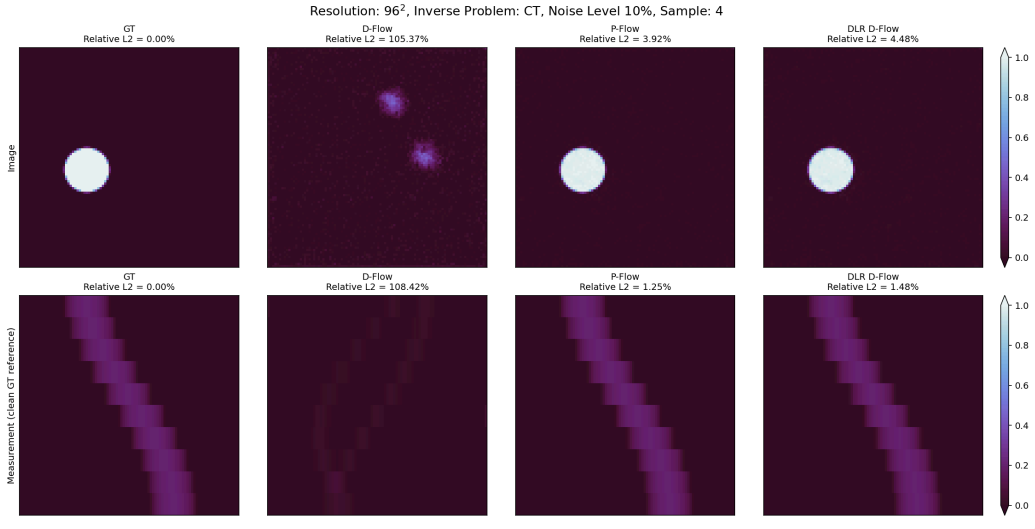
Example 2 · $\Delta = 111.19$ percentage points · CT · 96^2 · Sample 9

D-Flow: **114.96%** · P-Flow: **2.83%** · DLR D-Flow: **3.77%**



Example 3 · $\Delta = 100.89$ percentage points · CT · 96^2 · Sample 4

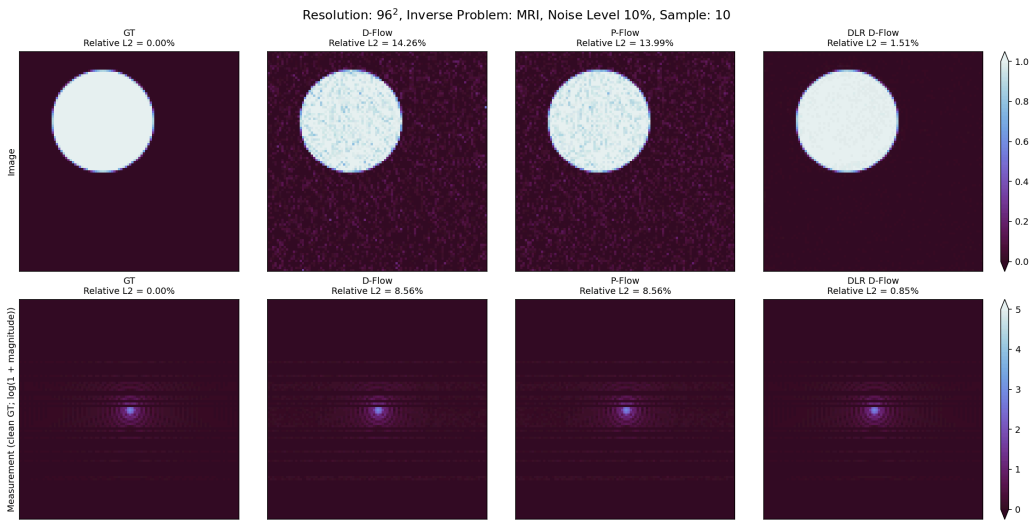
D-Flow: **105.37%** · P-Flow: **3.92%** · DLR D-Flow: **4.48%**



▼ MRI · 20/30 qualifying cases · showing 3 · 10% noise

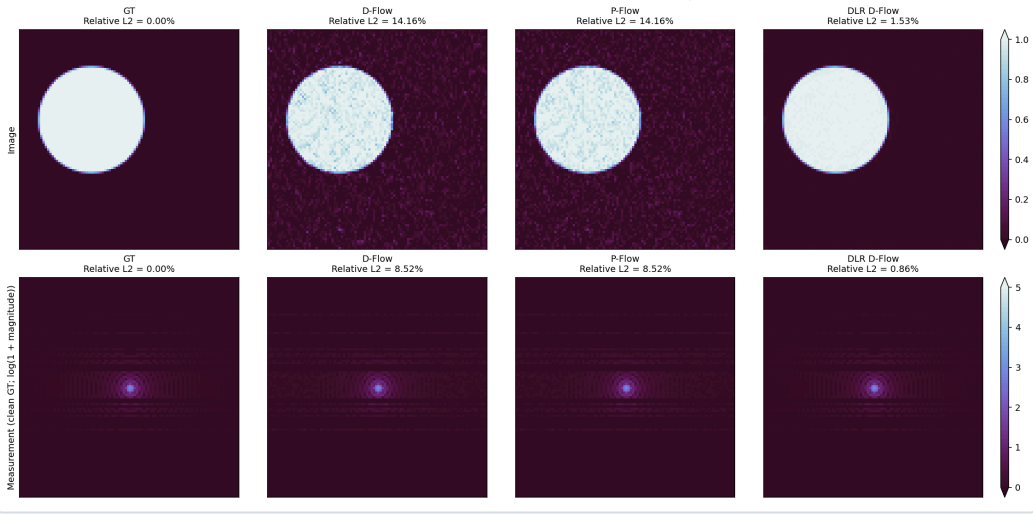
Example 1 · $\Delta = 12.76$ percentage points · MRI · 96^2 · Sample 10

D-Flow: **14.26%** · P-Flow: **13.99%** · DLR D-Flow: **1.51%**



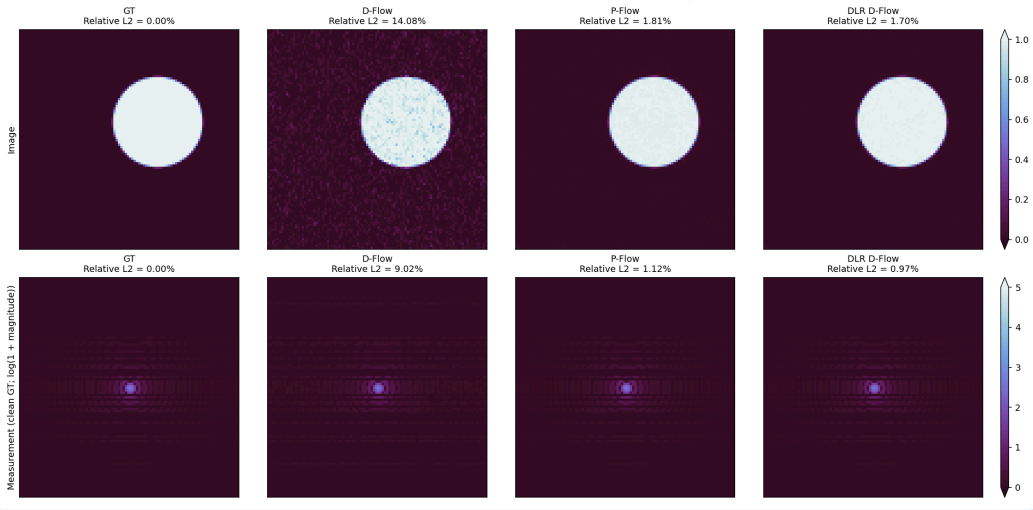
Example 2 · $\Delta = 12.64$ percentage points · MRI · 96^2 · Sample 11

D-Flow: **14.16%** · P-Flow: **14.16%** · DLR D-Flow: **1.53%**



Example 3 · $\Delta = 12.39$ percentage points · MRI · 96^2 · Sample 12

D-Flow: **14.08%** · P-Flow: **1.81%** · DLR D-Flow: **1.70%**

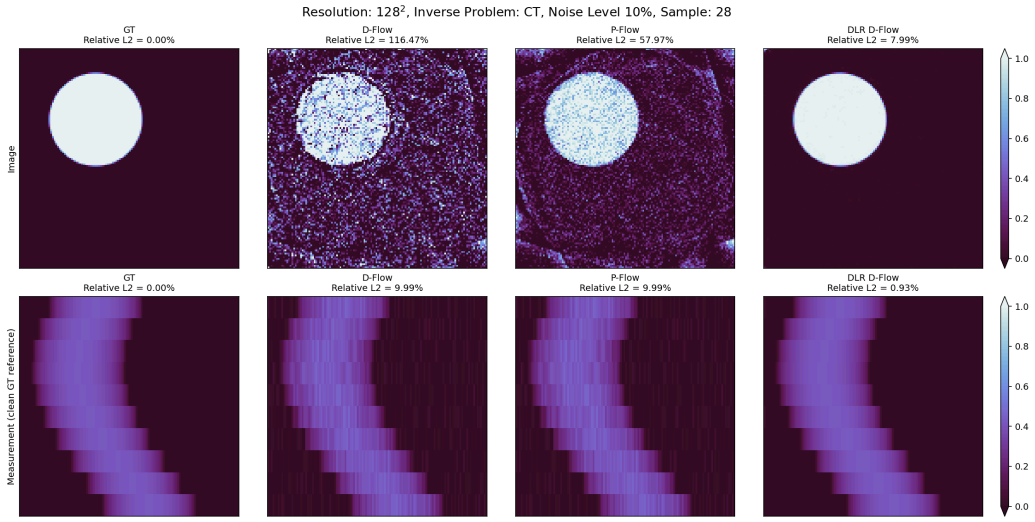


▼ Resolution: 128^2

▼ CT · 25/30 qualifying cases · showing 3 · 10% noise

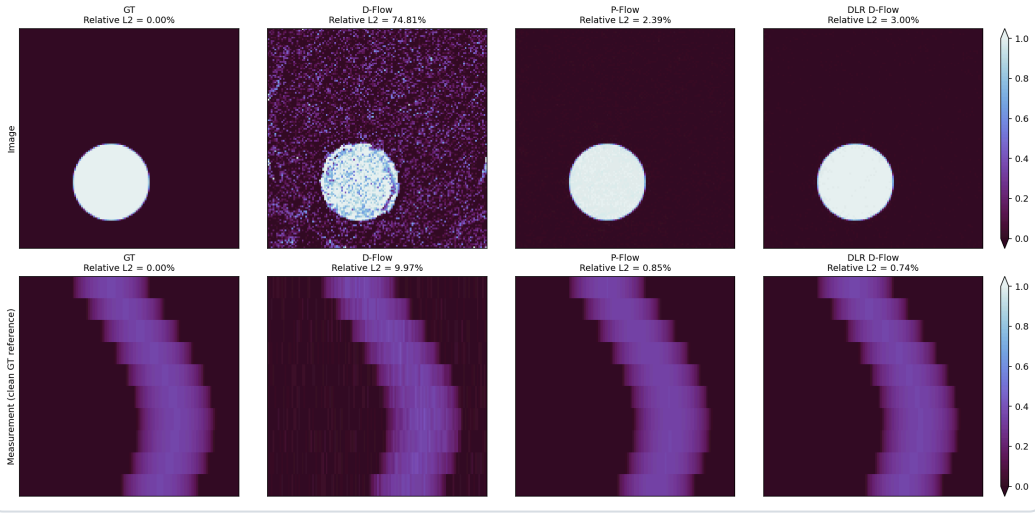
Example 1 · $\Delta = 108.48$ percentage points · CT · 128^2 · Sample 28

D-Flow: **116.47%** · P-Flow: **57.97%** · DLR D-Flow: **7.99%**



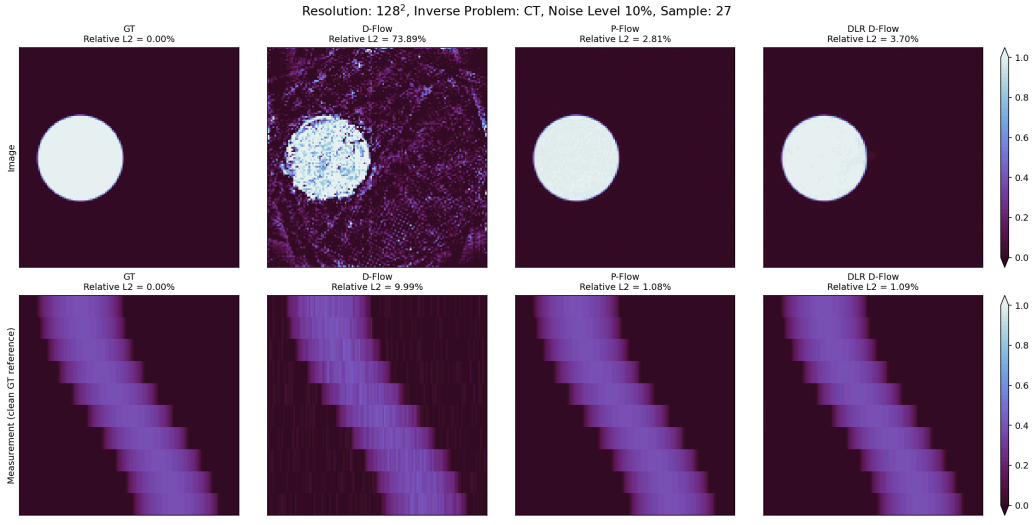
Example 2 · $\Delta = 71.80$ percentage points · CT · 128^2 · Sample 21

D-Flow: **74.81%** · P-Flow: **2.39%** · DLR D-Flow: **3.00%**



Example 3 · $\Delta = 70.19$ percentage points · CT · 128^2 · Sample 27

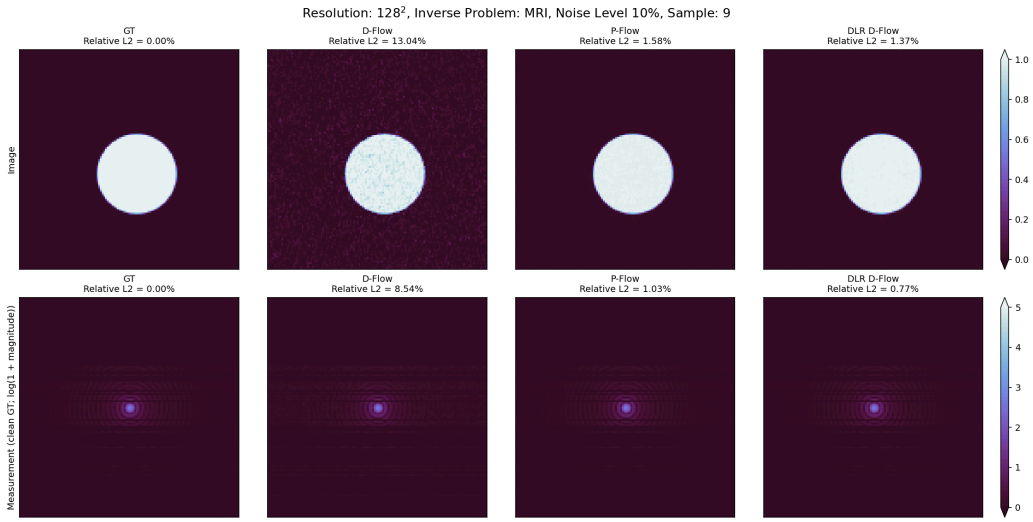
D-Flow: **73.89%** · P-Flow: **2.81%** · DLR D-Flow: **3.70%**



▼ MRI · 24/30 qualifying cases · showing 3 · 10% noise

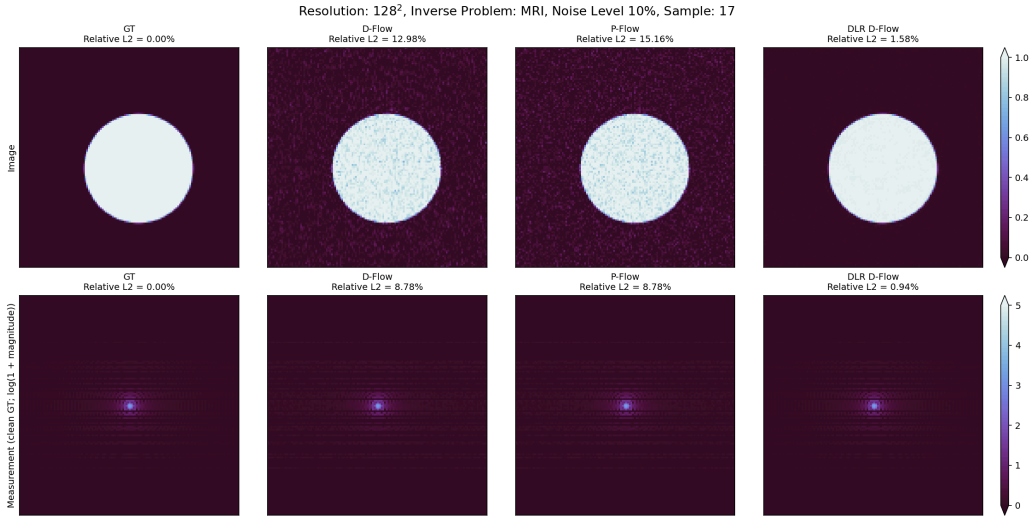
Example 1 · $\Delta = 11.67$ percentage points · MRI · 128^2 · Sample 9

D-Flow: **13.04%** · P-Flow: **1.58%** · DLR D-Flow: **1.37%**



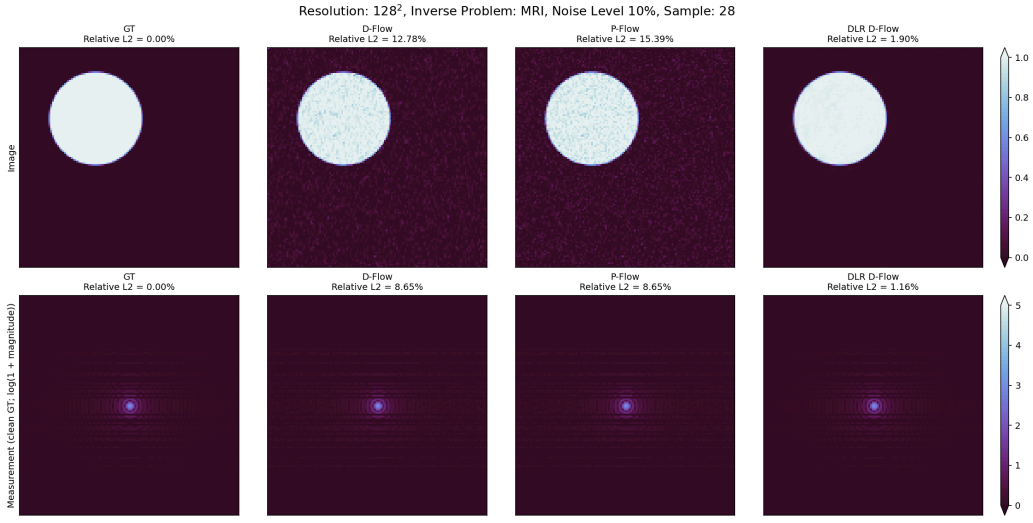
Example 2 · $\Delta = 11.40$ percentage points · MRI · 128^2 · Sample 17

D-Flow: **12.98%** · P-Flow: **15.16%** · DLR D-Flow: **1.58%**



Example 3 · $\Delta = 10.88$ percentage points · MRI · 128^2 · Sample 28

D-Flow: **12.78%** · P-Flow: **15.39%** · DLR D-Flow: **1.90%**

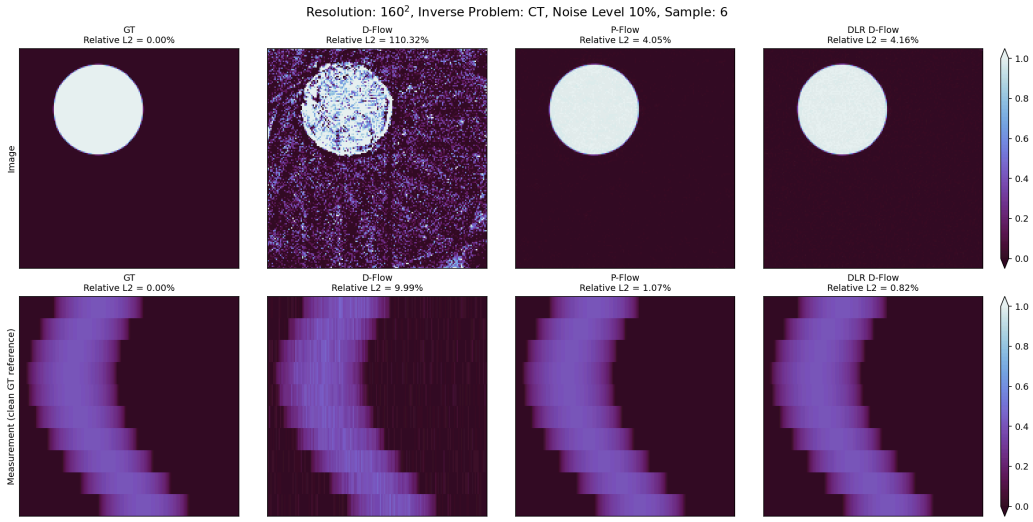


▼ Resolution: 160^2

▼ CT · 22/30 qualifying cases · showing 3 · 10% noise

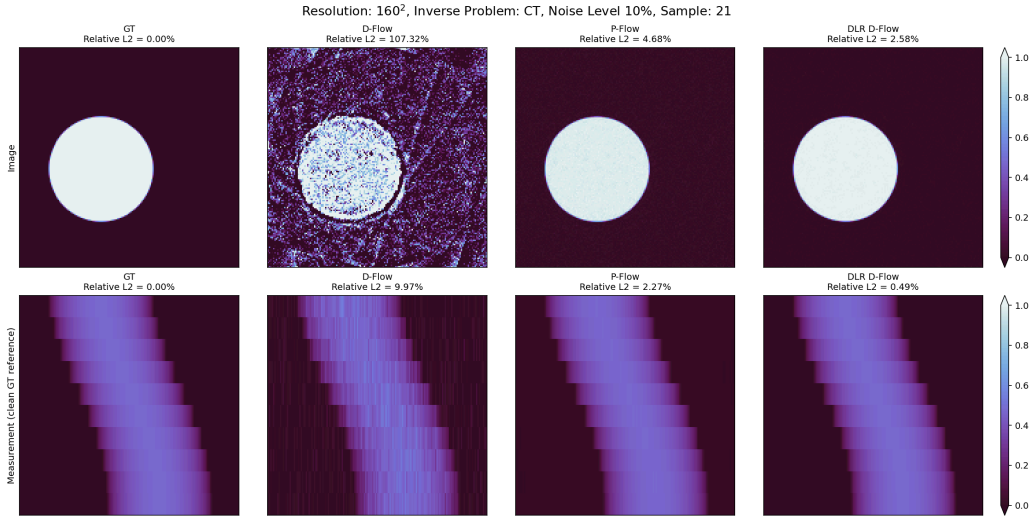
Example 1 · $\Delta = 106.16$ percentage points · CT · 160^2 · Sample 6

D-Flow: **110.32%** · P-Flow: **4.05%** · DLR D-Flow: **4.16%**



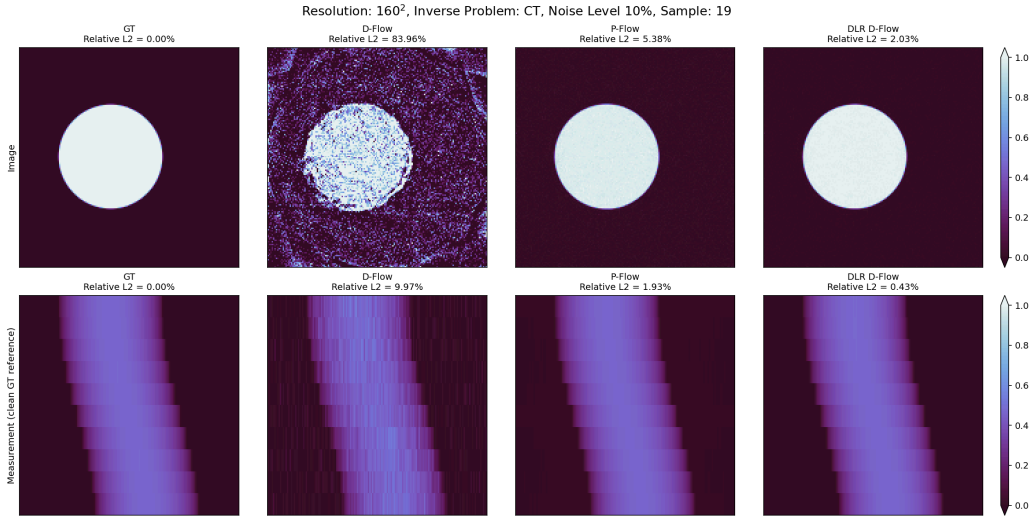
Example 2 · $\Delta = 104.74$ percentage points · CT · 160^2 · Sample 21

D-Flow: **107.32%** · P-Flow: **4.68%** · DLR D-Flow: **2.58%**



Example 3 · $\Delta = 81.92$ percentage points · CT · 160^2 · Sample 19

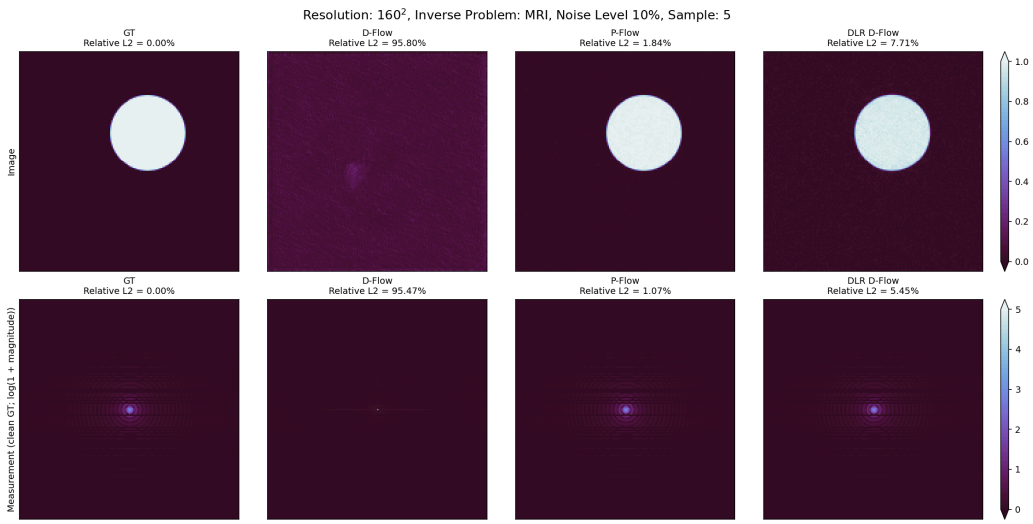
D-Flow: **83.96%** · P-Flow: **5.38%** · DLR D-Flow: **2.03%**



▼ MRI · 18/30 qualifying cases · showing 3 · 10% noise

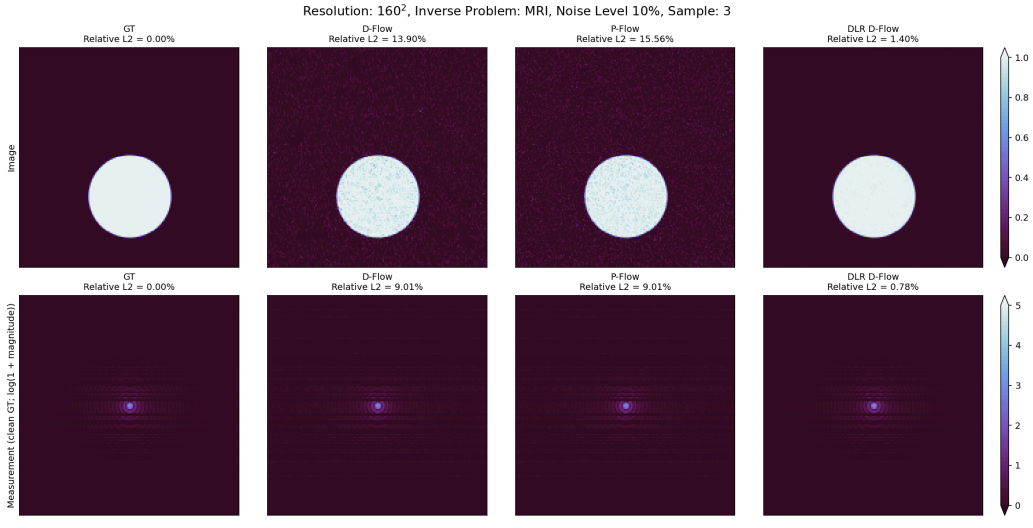
Example 1 · $\Delta = 88.09$ percentage points · MRI · 160^2 · Sample 5

D-Flow: **95.80%** · P-Flow: **1.84%** · DLR D-Flow: **7.71%**



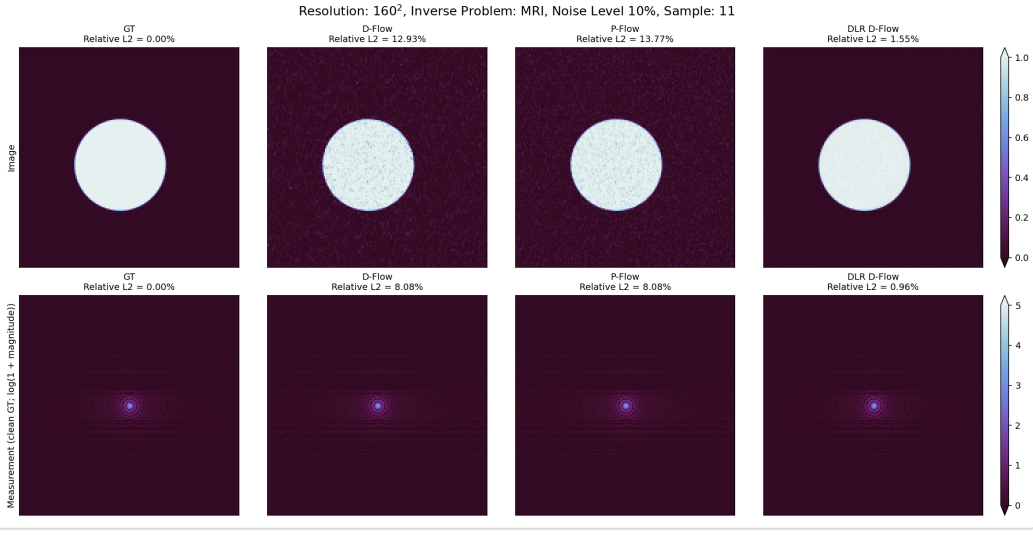
Example 2 · $\Delta = 12.50$ percentage points · MRI · 160^2 · Sample 3

D-Flow: **13.90%** · P-Flow: **15.56%** · DLR D-Flow: **1.40%**



Example 3 · $\Delta = 11.39$ percentage points · MRI · 160^2 · Sample 11

D-Flow: **12.93%** · P-Flow: **13.77%** · DLR D-Flow: **1.55%**

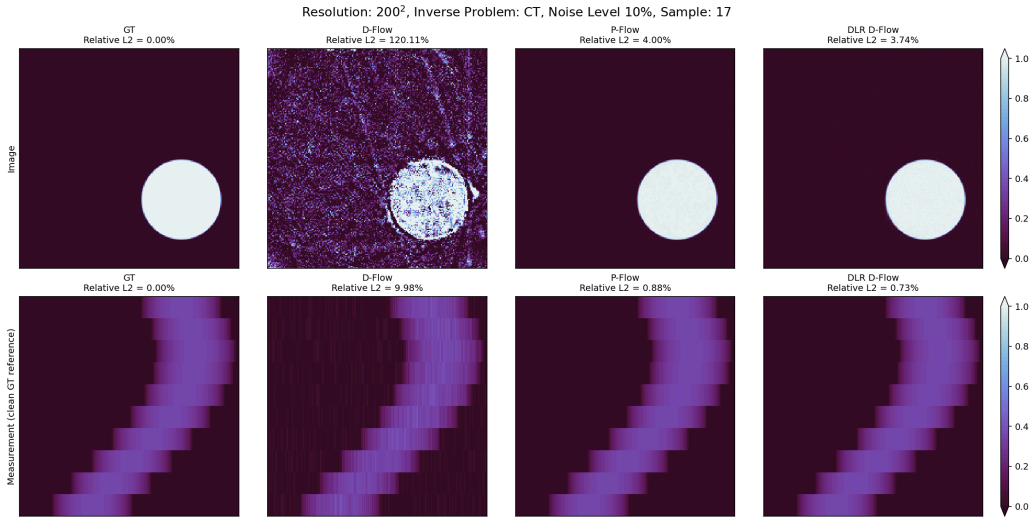


▼ Resolution: 200^2

▼ CT · 24/30 qualifying cases · showing 3 · 10% noise

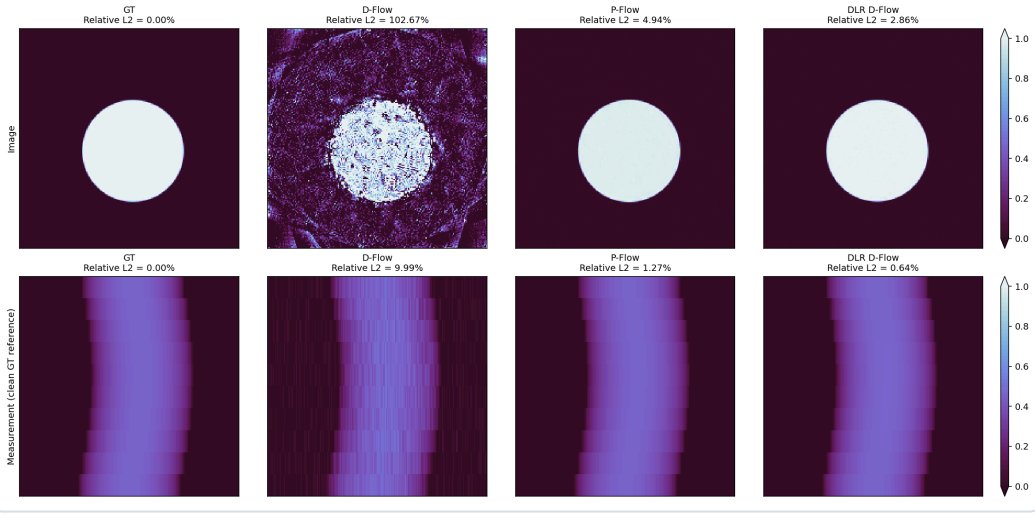
Example 1 · $\Delta = 116.38$ percentage points · CT · 200^2 · Sample 17

D-Flow: **120.11%** · P-Flow: **4.00%** · DLR D-Flow: **3.74%**



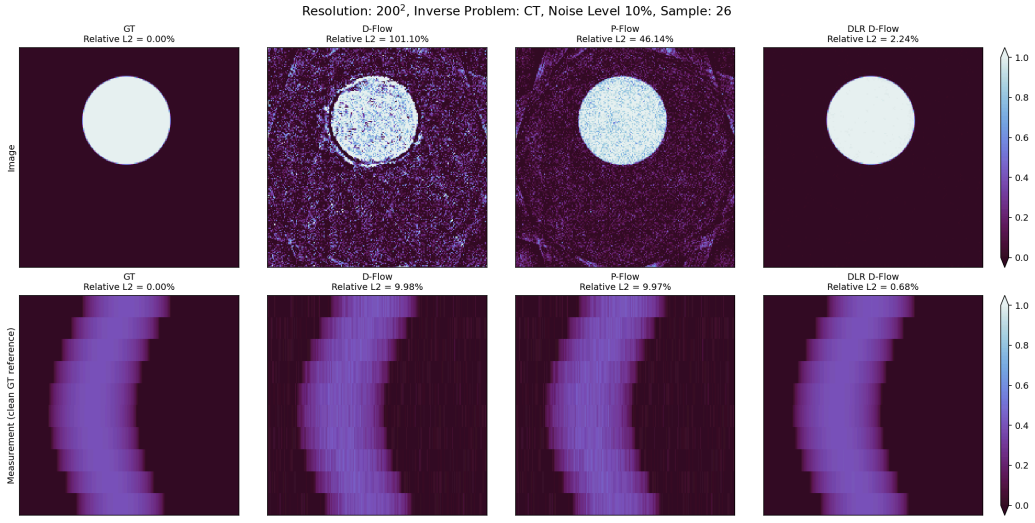
Example 2 · $\Delta = 99.81$ percentage points · CT · 200^2 · Sample 0

D-Flow: **102.67%** · P-Flow: **4.94%** · DLR D-Flow: **2.86%**



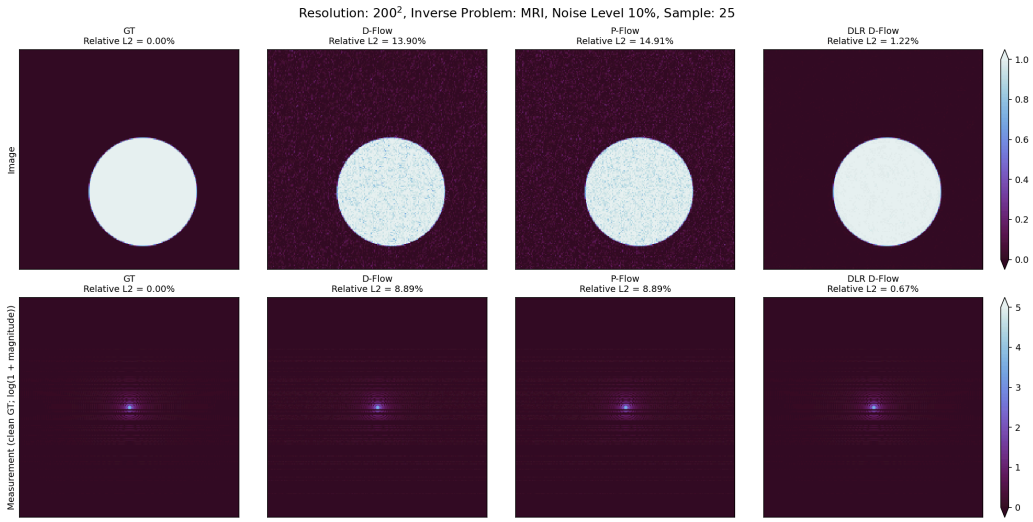
Example 3 · $\Delta = 98.87$ percentage points · CT · 200^2 · Sample 26

D-Flow: **101.10%** · P-Flow: **46.14%** · DLR D-Flow: **2.24%**

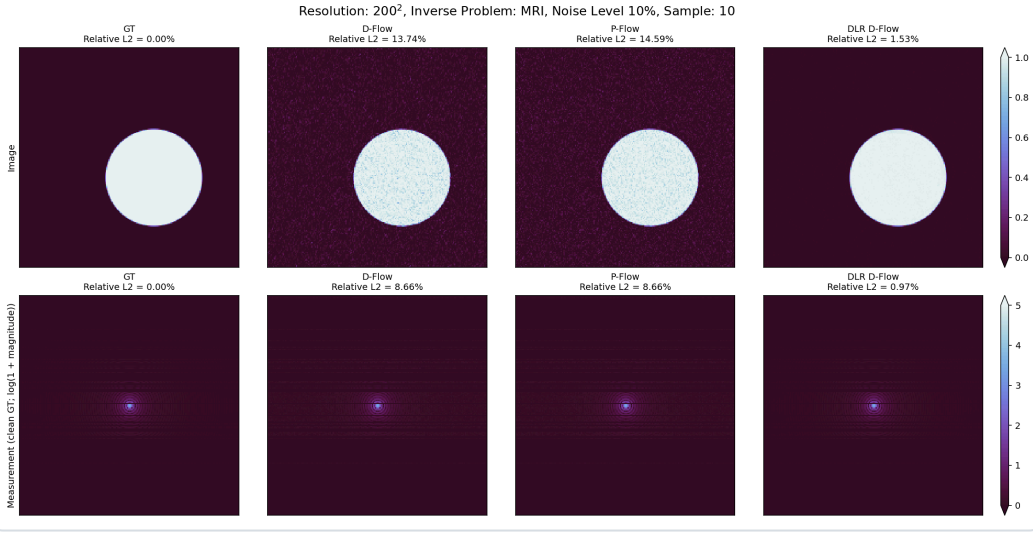


▼ MRI · 19/30 qualifying cases · showing 3 · 10% noise

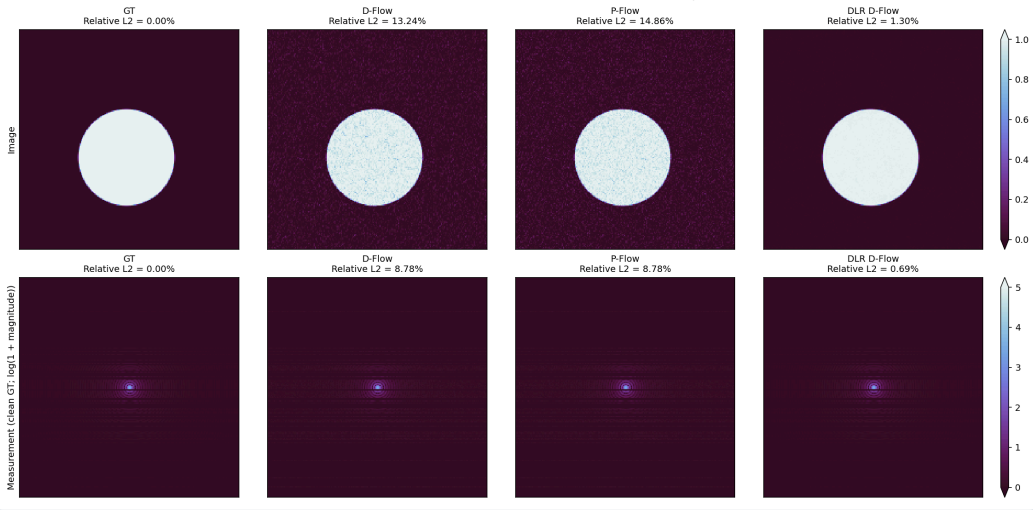
Example 1 · $\Delta = 12.68$ percentage points · MRI · 200^2 · Sample 25
D-Flow: **13.90%** · P-Flow: **14.91%** · DLR D-Flow: **1.22%**



Example 2 · $\Delta = 12.21$ percentage points · MRI · 200^2 · Sample 10
D-Flow: **13.74%** · P-Flow: **14.59%** · DLR D-Flow: **1.53%**



Example 3 · $\Delta = 11.94$ percentage points · MRI · 200^2 · Sample 18
D-Flow: **13.24%** · P-Flow: **14.86%** · DLR D-Flow: **1.30%**

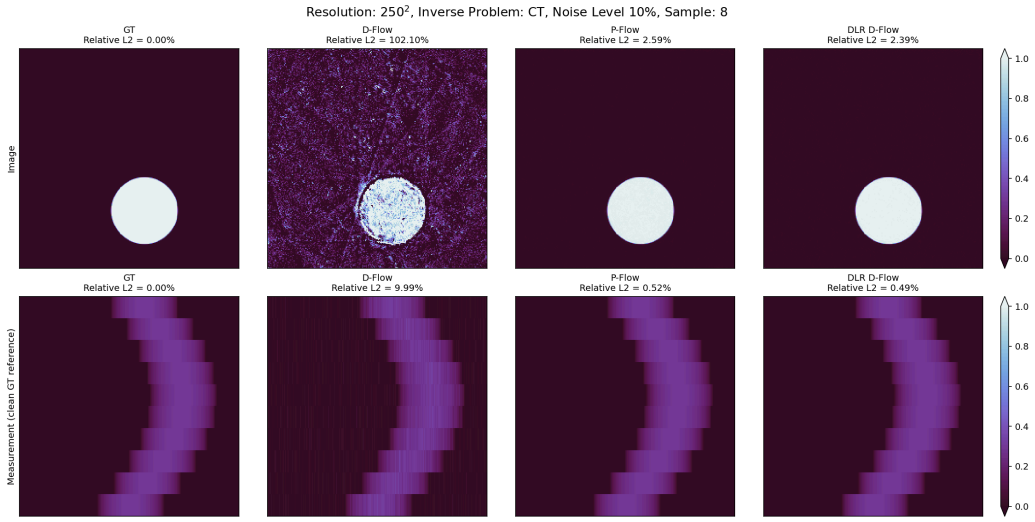


▼ Resolution: 250^2

▼ CT · 22/30 qualifying cases · showing 3 · 10% noise

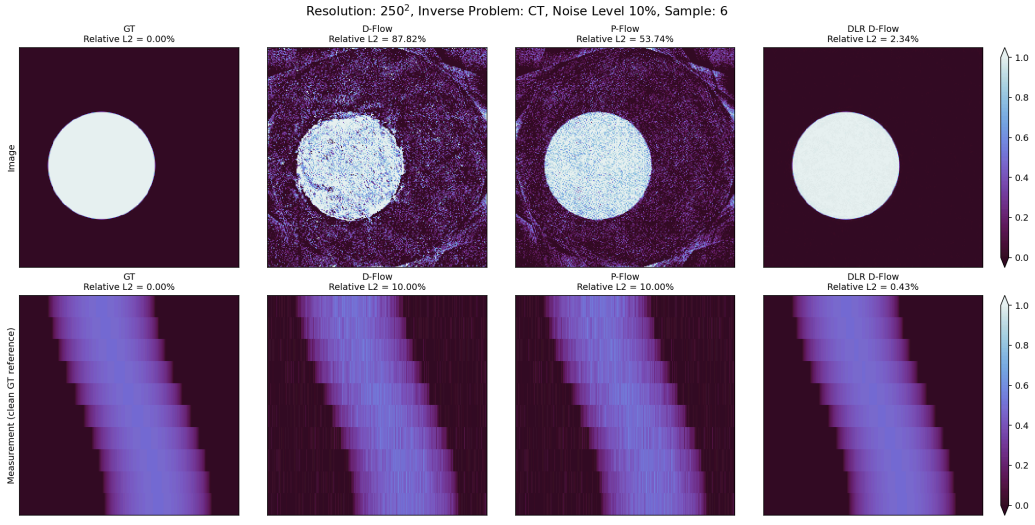
Example 1 · $\Delta = 99.71$ percentage points · CT · 250^2 · Sample 8

D-Flow: **102.10%** · P-Flow: **2.59%** · DLR D-Flow: **2.39%**



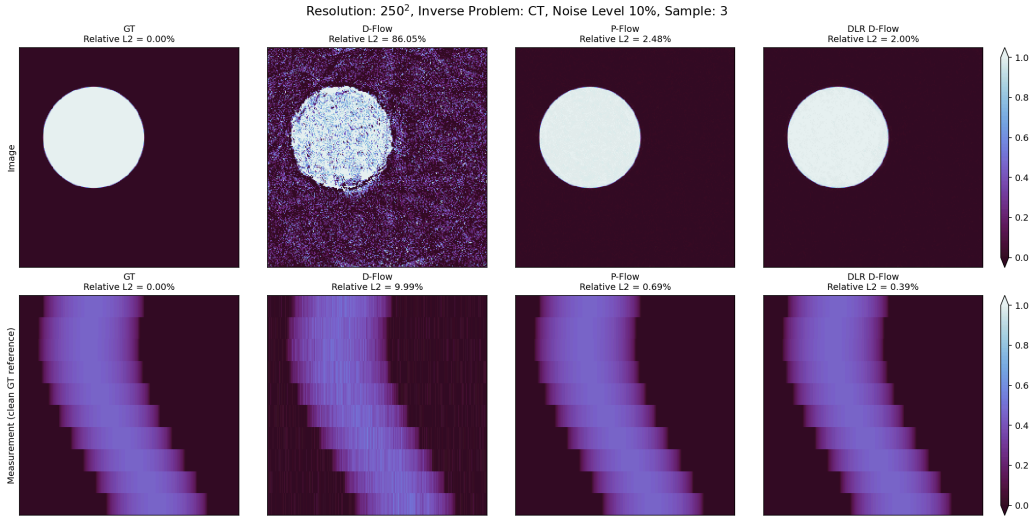
Example 2 · $\Delta = 85.48$ percentage points · CT · 250^2 · Sample 6

D-Flow: **87.82%** · P-Flow: **53.74%** · DLR D-Flow: **2.34%**



Example 3 · $\Delta = 84.05$ percentage points · CT · 250^2 · Sample 3

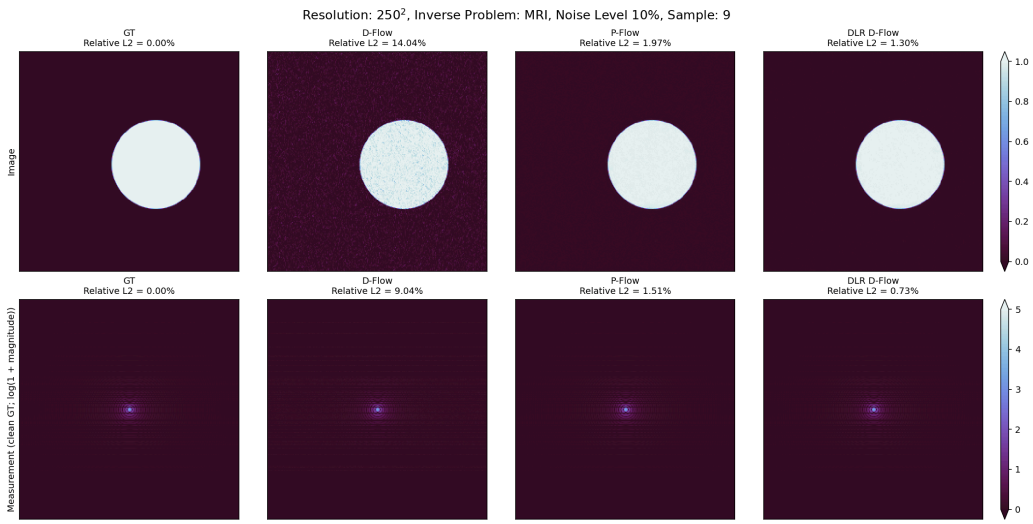
D-Flow: **86.05%** · P-Flow: **2.48%** · DLR D-Flow: **2.00%**



▼ MRI · 21/30 qualifying cases · showing 3 · 10% noise

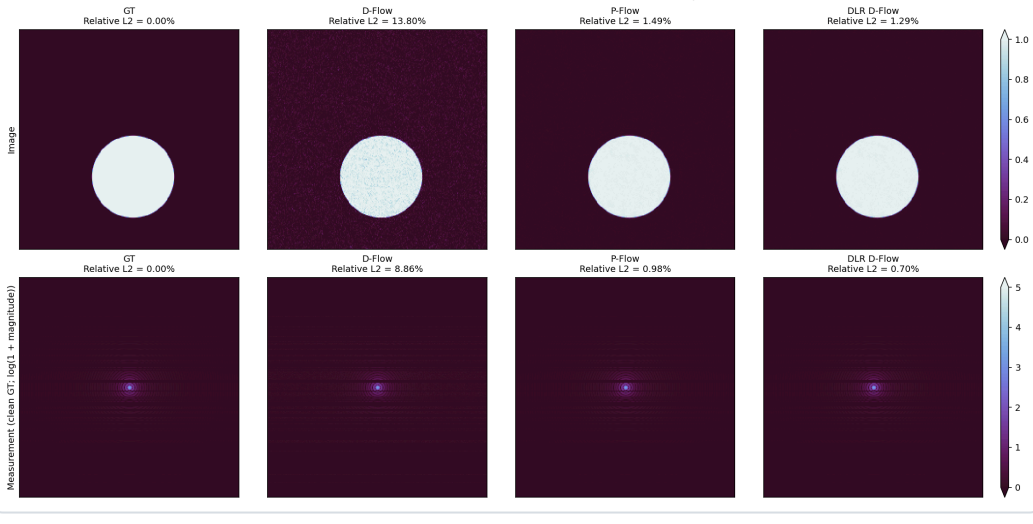
Example 1 · $\Delta = 12.74$ percentage points · MRI · 250^2 · Sample 9

D-Flow: **14.04%** · P-Flow: **1.97%** · DLR D-Flow: **1.30%**



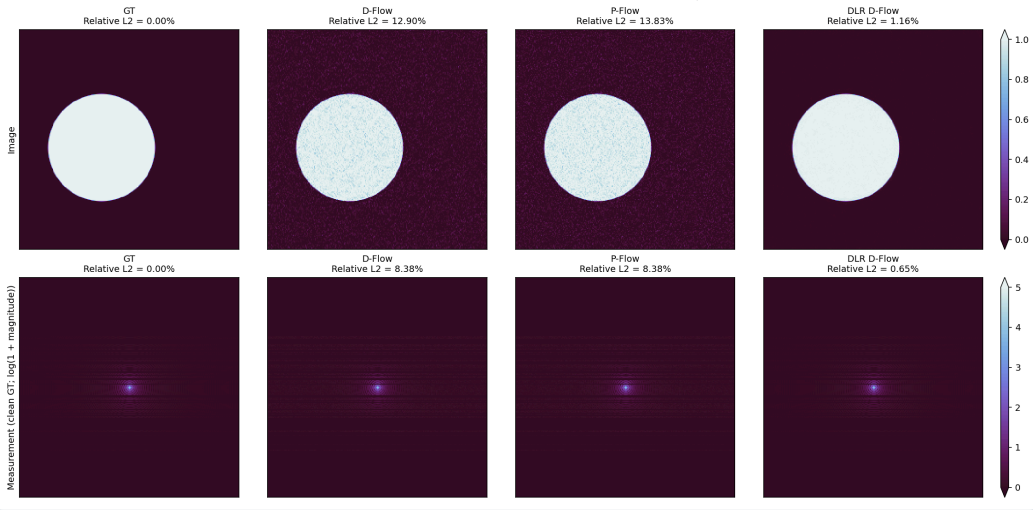
Example 2 · $\Delta = 12.50$ percentage points · MRI · 250^2 · Sample 20

D-Flow: **13.80%** · P-Flow: **1.49%** · DLR D-Flow: **1.29%**



Example 3 · $\Delta = 11.75$ percentage points · MRI · 250^2 · Sample 6

D-Flow: **12.90%** · P-Flow: **13.83%** · DLR D-Flow: **1.16%**

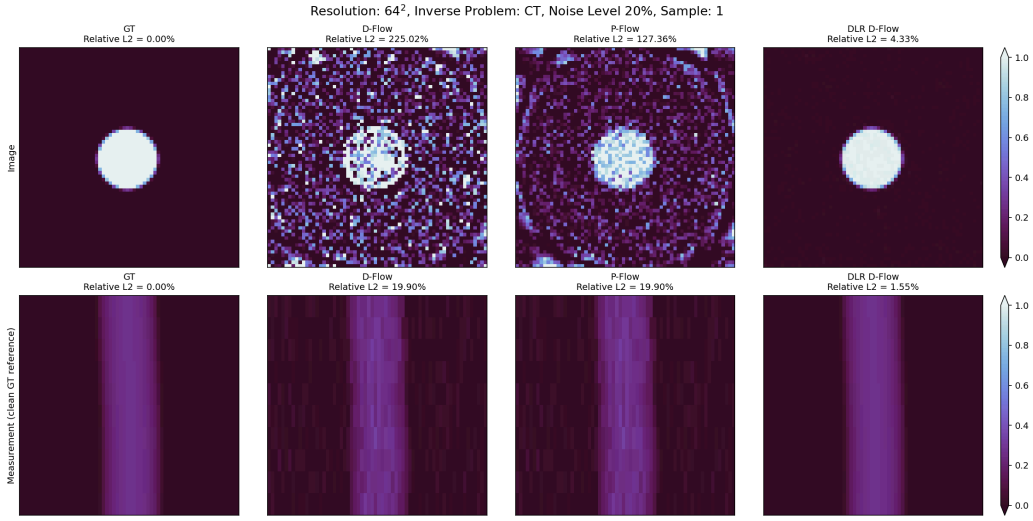


▼ 20% noise · DLR advantage by resolution

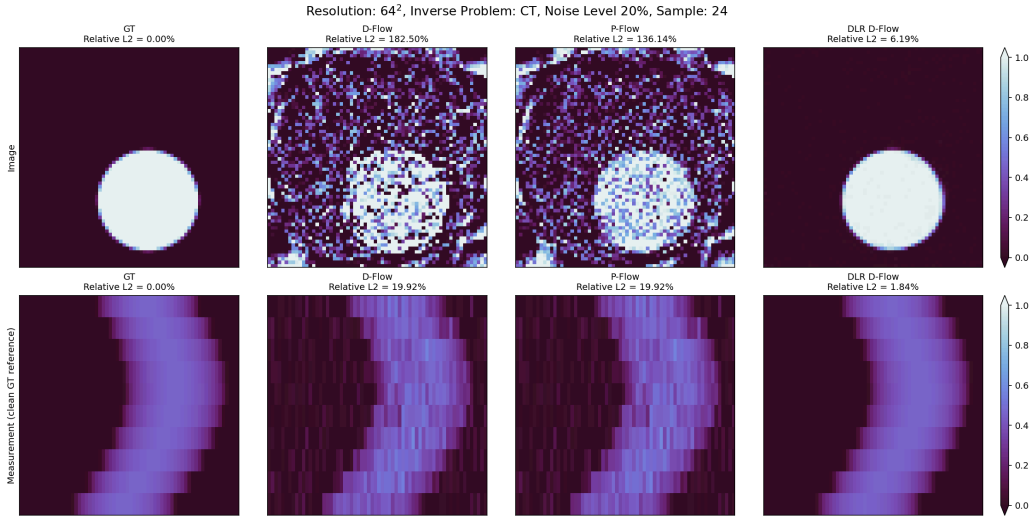
▼ Resolution: 64^2

▼ CT · 22/30 qualifying cases · showing 3 · 20% noise

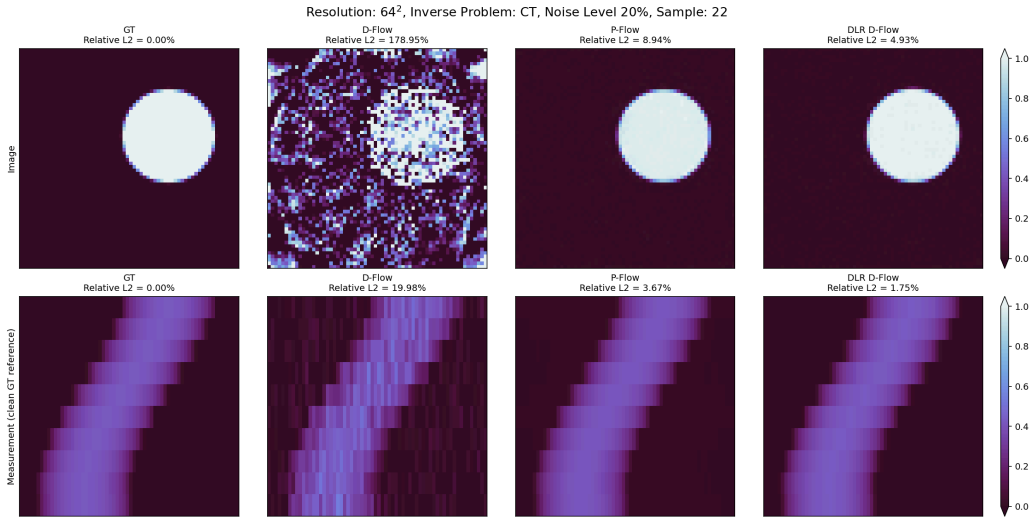
Example 1 · $\Delta = 220.69$ percentage points · CT · 64^2 · Sample 1
D-Flow: **225.02%** · P-Flow: **127.36%** · DLR D-Flow: **4.33%**



Example 2 · $\Delta = 176.31$ percentage points · CT · 64^2 · Sample 24
D-Flow: **182.50%** · P-Flow: **136.14%** · DLR D-Flow: **6.19%**



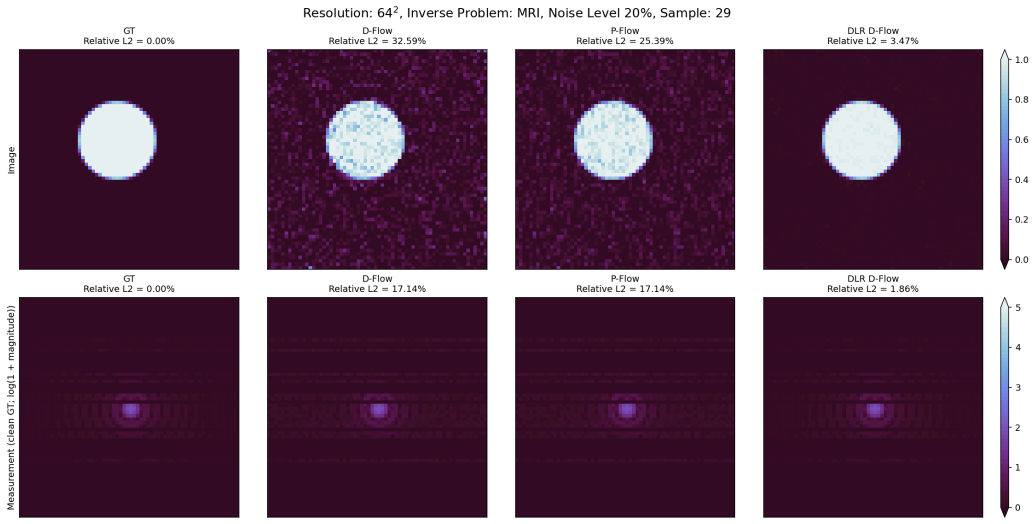
Example 3 · $\Delta = 174.02$ percentage points · CT · 64^2 · Sample 22
D-Flow: **178.95%** · P-Flow: **8.94%** · DLR D-Flow: **4.93%**



▼ MRI · 25/30 qualifying cases · showing 3 · 20% noise

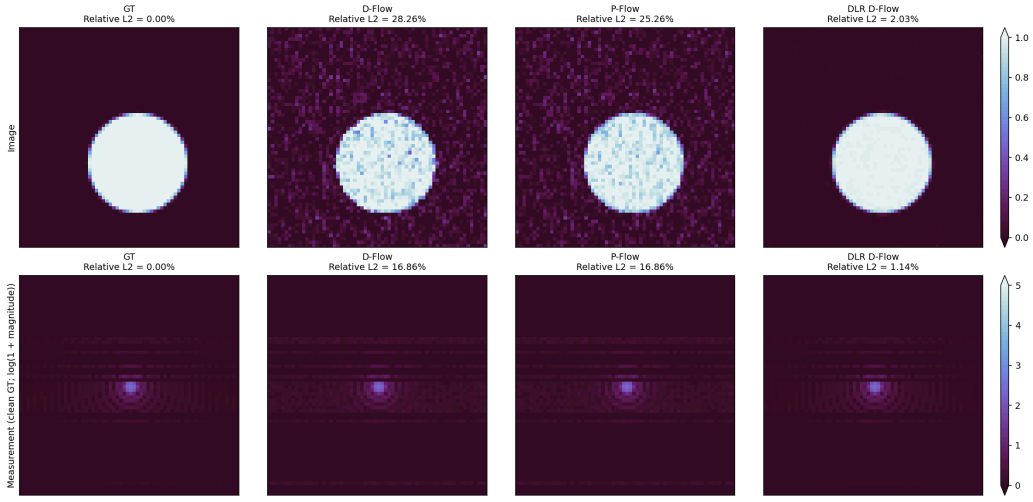
Example 1 · $\Delta = 29.12$ percentage points · MRI · 64^2 · Sample 29

D-Flow: **32.59%** · P-Flow: **25.39%** · DLR D-Flow: **3.47%**



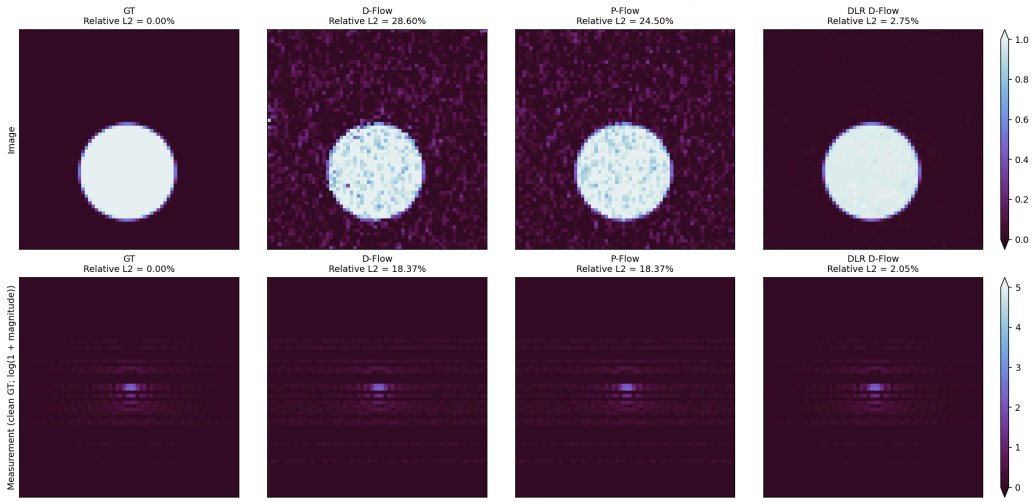
Example 2 · $\Delta = 26.23$ percentage points · MRI · 64^2 · Sample 20

D-Flow: **28.26%** · P-Flow: **25.26%** · DLR D-Flow: **2.03%**



Example 3 · $\Delta = 25.85$ percentage points · MRI · 64^2 · Sample 5

D-Flow: **28.60%** · P-Flow: **24.50%** · DLR D-Flow: **2.75%**

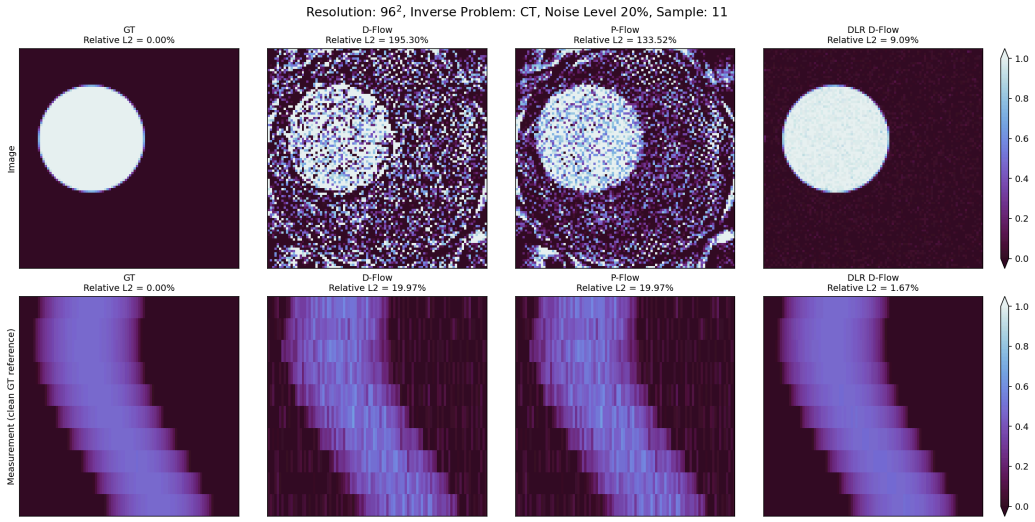


▼ Resolution: 96^2

▼ CT · 19/30 qualifying cases · showing 3 · 20% noise

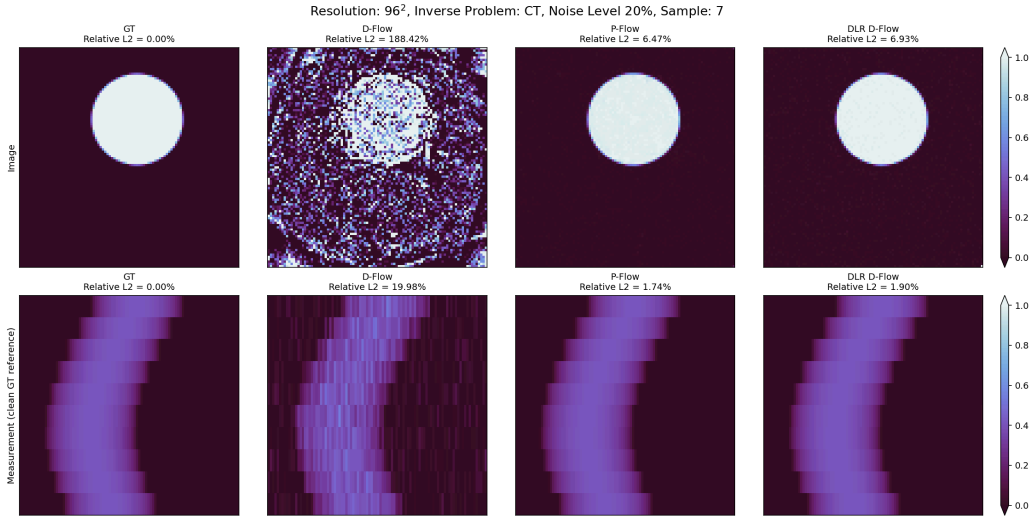
Example 1 · $\Delta = 186.22$ percentage points · CT · 96^2 · Sample 11

D-Flow: **195.30%** · P-Flow: **133.52%** · DLR D-Flow: **9.09%**



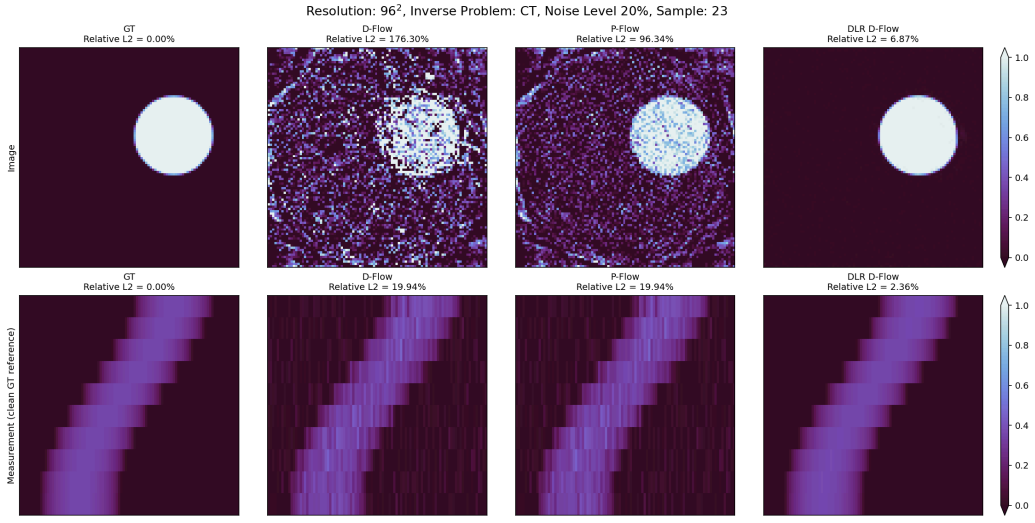
Example 2 · $\Delta = 181.49$ percentage points · CT · 96^2 · Sample 7

D-Flow: **188.42%** · P-Flow: **6.47%** · DLR D-Flow: **6.93%**



Example 3 · $\Delta = 169.44$ percentage points · CT · 96^2 · Sample 23

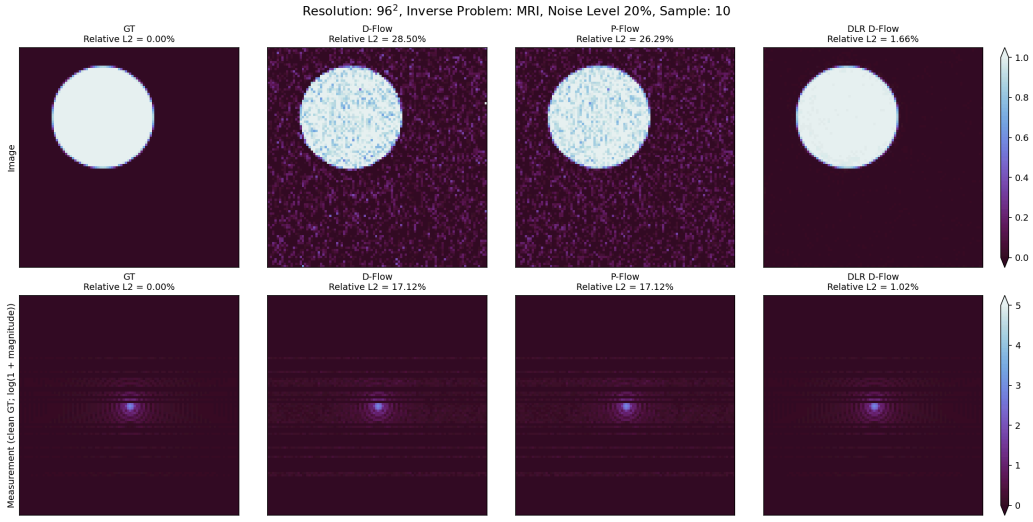
D-Flow: **176.30%** · P-Flow: **96.34%** · DLR D-Flow: **6.87%**



▼ MRI · 20/30 qualifying cases · showing 3 · 20% noise

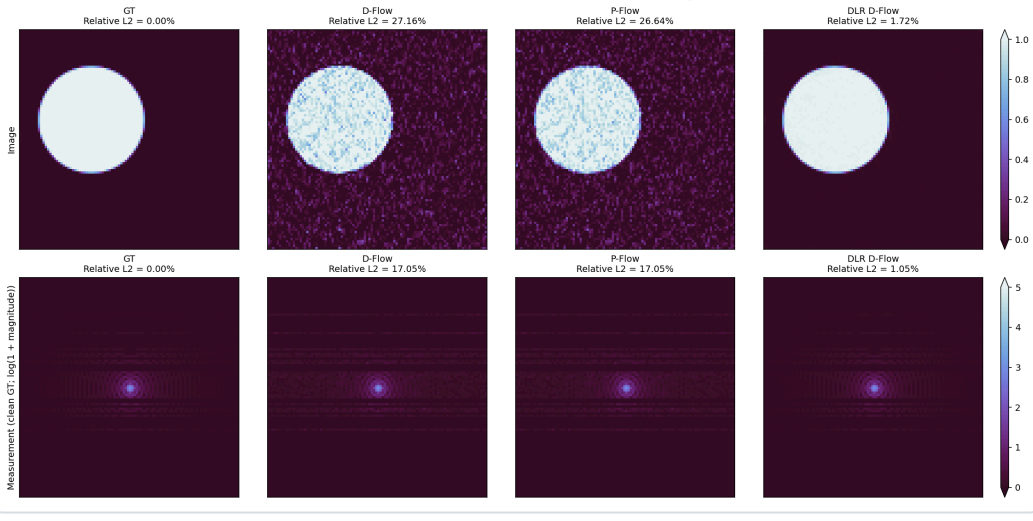
Example 1 · $\Delta = 26.84$ percentage points · MRI · 96^2 · Sample 10

D-Flow: **28.50%** · P-Flow: **26.29%** · DLR D-Flow: **1.66%**



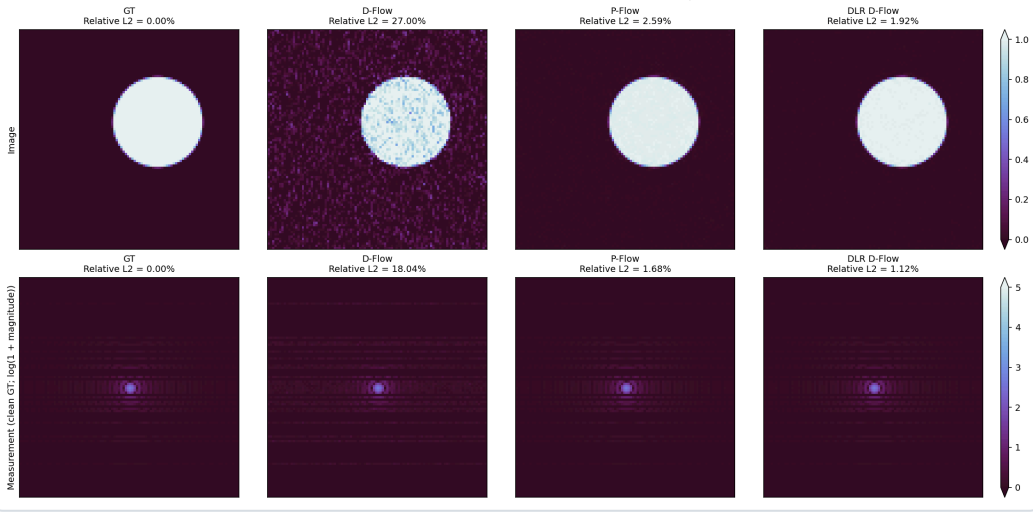
Example 2 · $\Delta = 25.44$ percentage points · MRI · 96^2 · Sample 11

D-Flow: **27.16%** · P-Flow: **26.64%** · DLR D-Flow: **1.72%**



Example 3 · $\Delta = 25.08$ percentage points · MRI · 96^2 · Sample 12

D-Flow: **27.00%** · P-Flow: **2.59%** · DLR D-Flow: **1.92%**

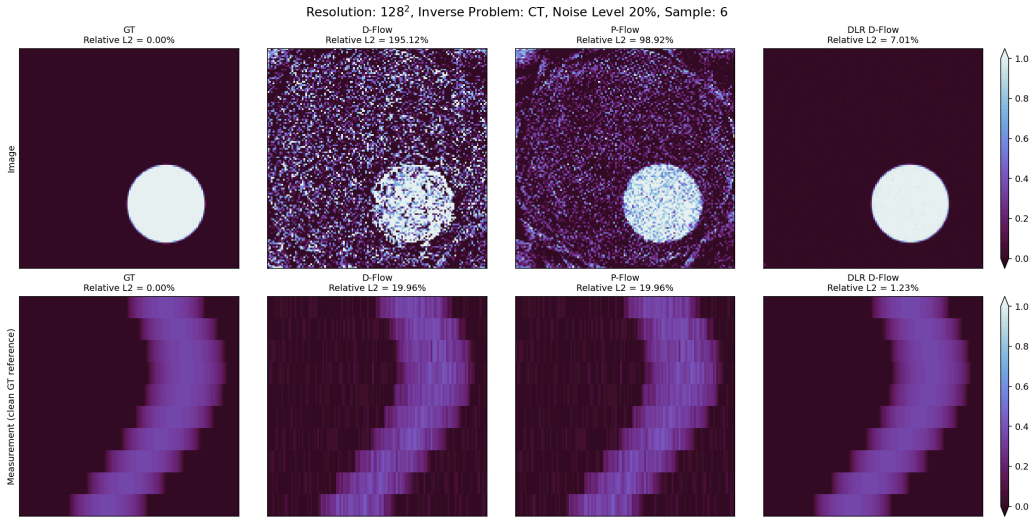


▼ Resolution: 128^2

▼ CT · 22/30 qualifying cases · showing 3 · 20% noise

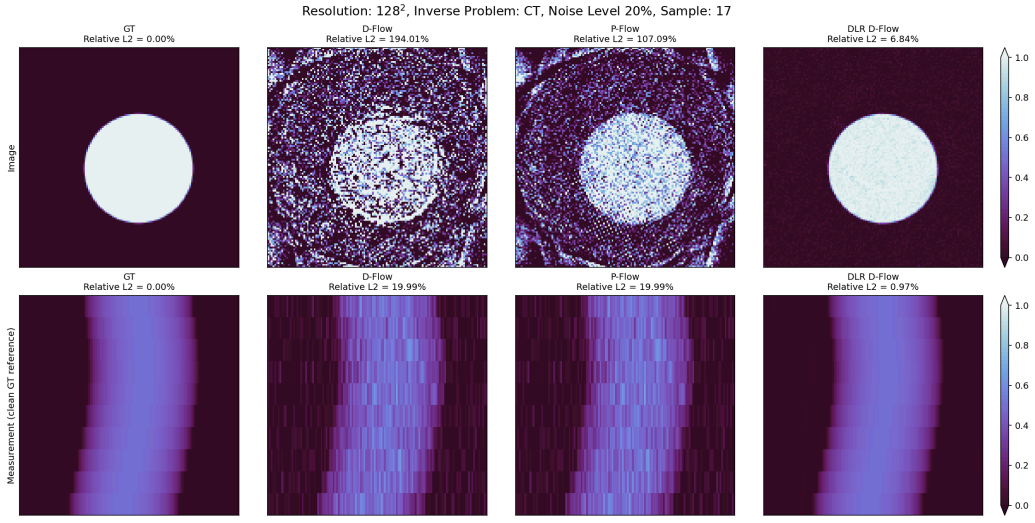
Example 1 · $\Delta = 188.11$ percentage points · CT · 128^2 · Sample 6

D-Flow: **195.12%** · P-Flow: **98.92%** · DLR D-Flow: **7.01%**



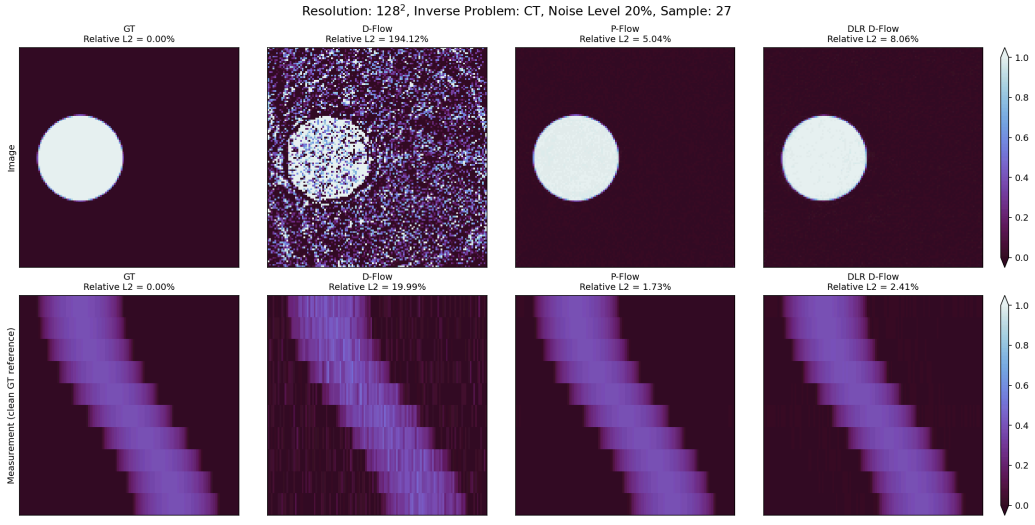
Example 2 · $\Delta = 187.16$ percentage points · CT · 128^2 · Sample 17

D-Flow: **194.01%** · P-Flow: **107.09%** · DLR D-Flow: **6.84%**



Example 3 · $\Delta = 186.06$ percentage points · CT · 128^2 · Sample 27

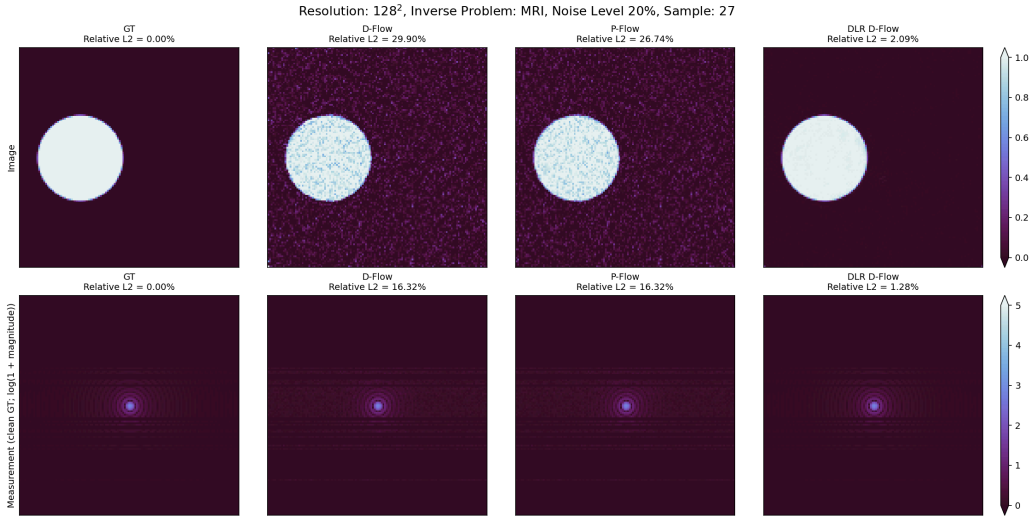
D-Flow: **194.12%** · P-Flow: **5.04%** · DLR D-Flow: **8.06%**



▼ MRI · 25/30 qualifying cases · showing 3 · 20% noise

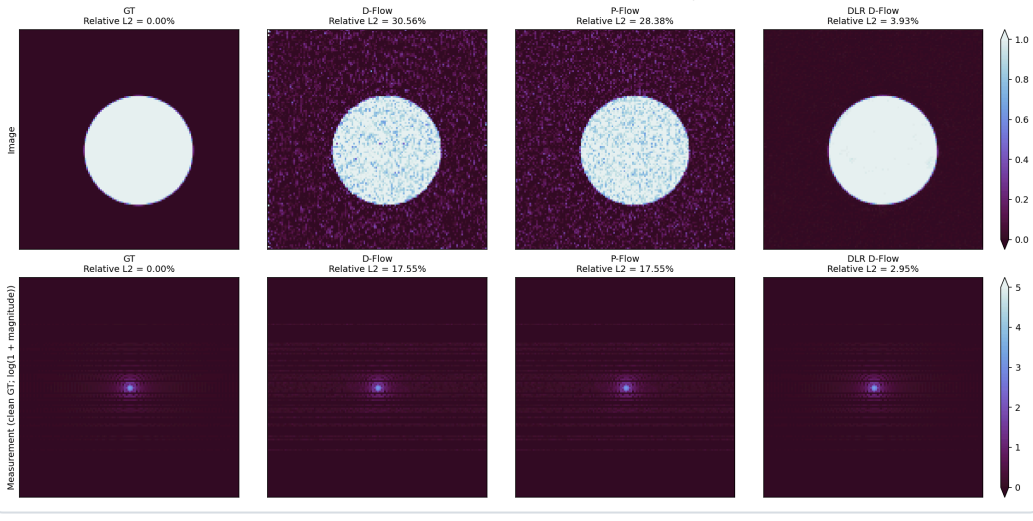
Example 1 · $\Delta = 27.81$ percentage points · MRI · 128^2 · Sample 27

D-Flow: **29.90%** · P-Flow: **26.74%** · DLR D-Flow: **2.09%**



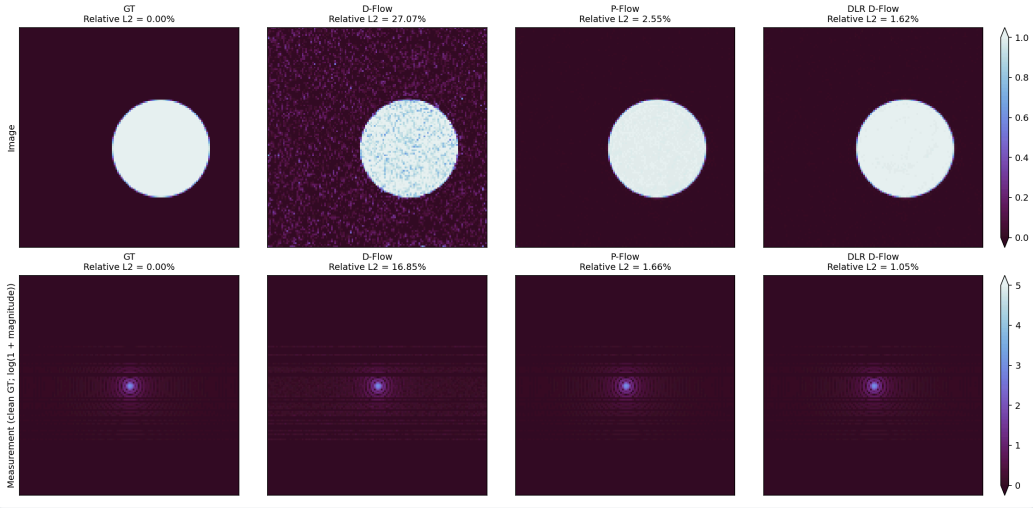
Example 2 · $\Delta = 26.63$ percentage points · MRI · 128^2 · Sample 17

D-Flow: **30.56%** · P-Flow: **28.38%** · DLR D-Flow: **3.93%**



Example 3 · $\Delta = 25.44$ percentage points · MRI · 128^2 · Sample 18

D-Flow: **27.07%** · P-Flow: **2.55%** · DLR D-Flow: **1.62%**

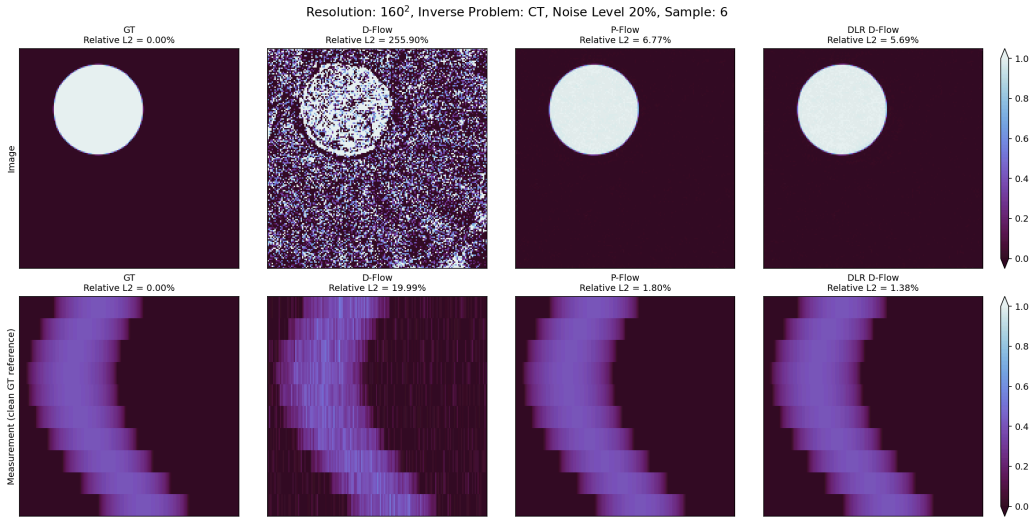


▼ Resolution: 160^2

▼ CT · 22/30 qualifying cases · showing 3 · 20% noise

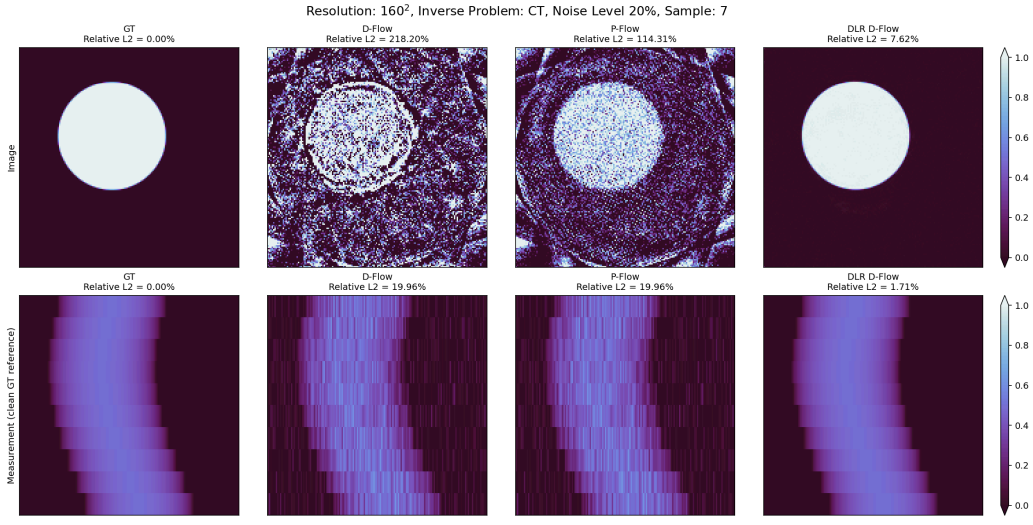
Example 1 · $\Delta = 250.21$ percentage points · CT · 160^2 · Sample 6

D-Flow: **255.90%** · P-Flow: **6.77%** · DLR D-Flow: **5.69%**



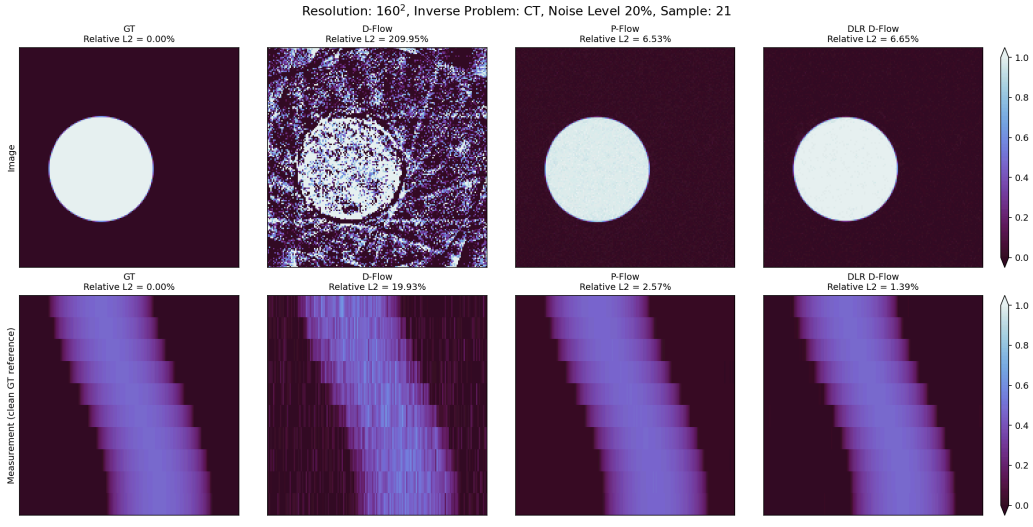
Example 2 · $\Delta = 210.59$ percentage points · CT · 160^2 · Sample 7

D-Flow: **218.20%** · P-Flow: **114.31%** · DLR D-Flow: **7.62%**



Example 3 · $\Delta = 203.31$ percentage points · CT · 160^2 · Sample 21

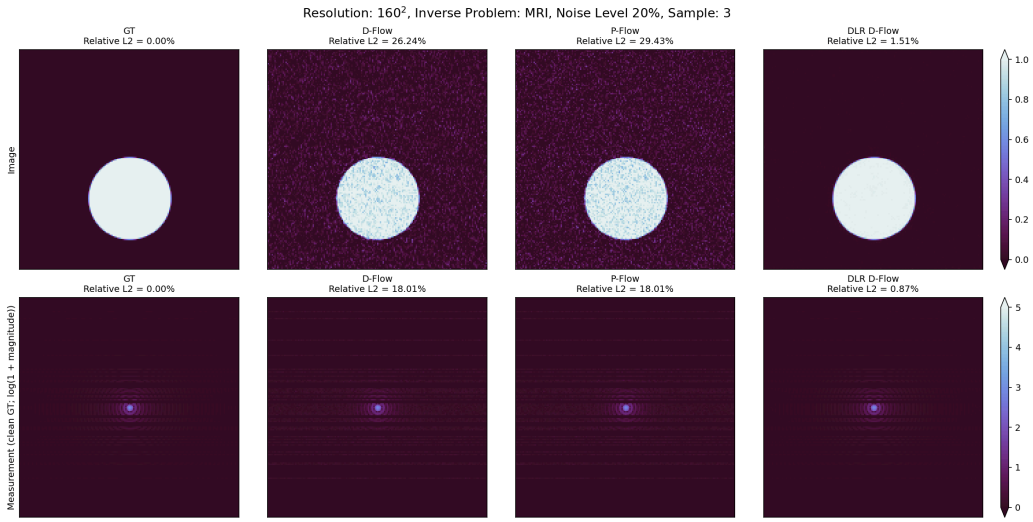
D-Flow: **209.95%** · P-Flow: **6.53%** · DLR D-Flow: **6.65%**



▼ MRI · 20/30 qualifying cases · showing 3 · 20% noise

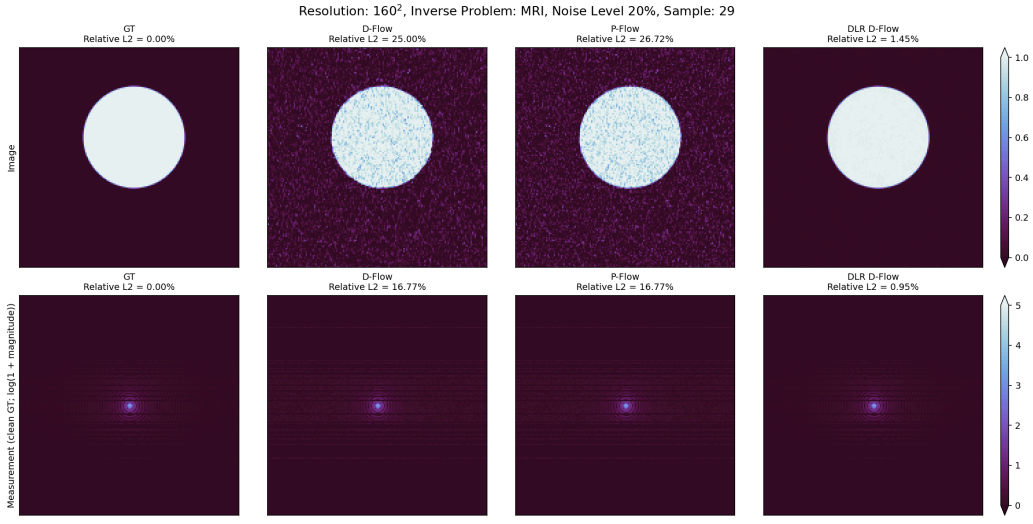
Example 1 · $\Delta = 24.73$ percentage points · MRI · 160^2 · Sample 3

D-Flow: **26.24%** · P-Flow: **29.43%** · DLR D-Flow: **1.51%**



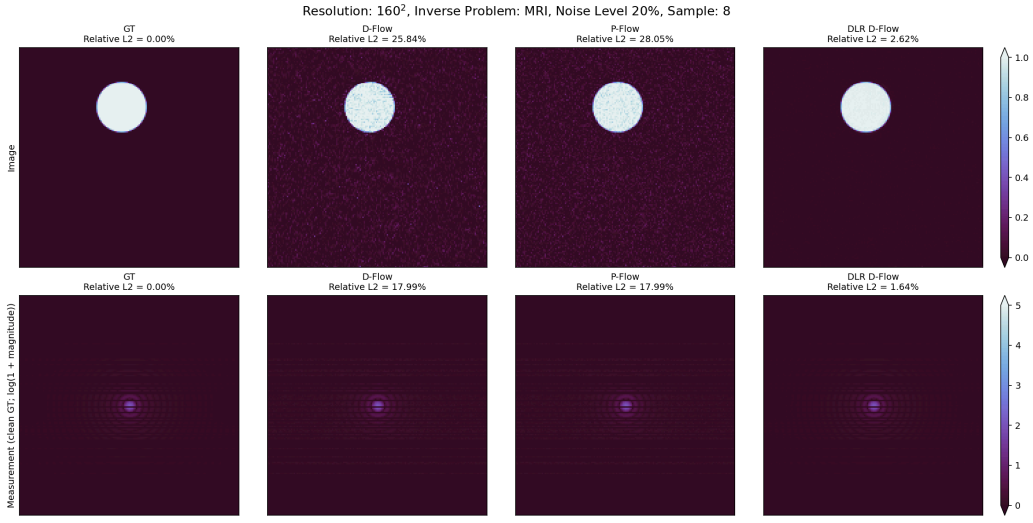
Example 2 · $\Delta = 23.55$ percentage points · MRI · 160^2 · Sample 29

D-Flow: **25.00%** · P-Flow: **26.72%** · DLR D-Flow: **1.45%**



Example 3 · $\Delta = 23.21$ percentage points · MRI · 160^2 · Sample 8

D-Flow: **25.84%** · P-Flow: **28.05%** · DLR D-Flow: **2.62%**

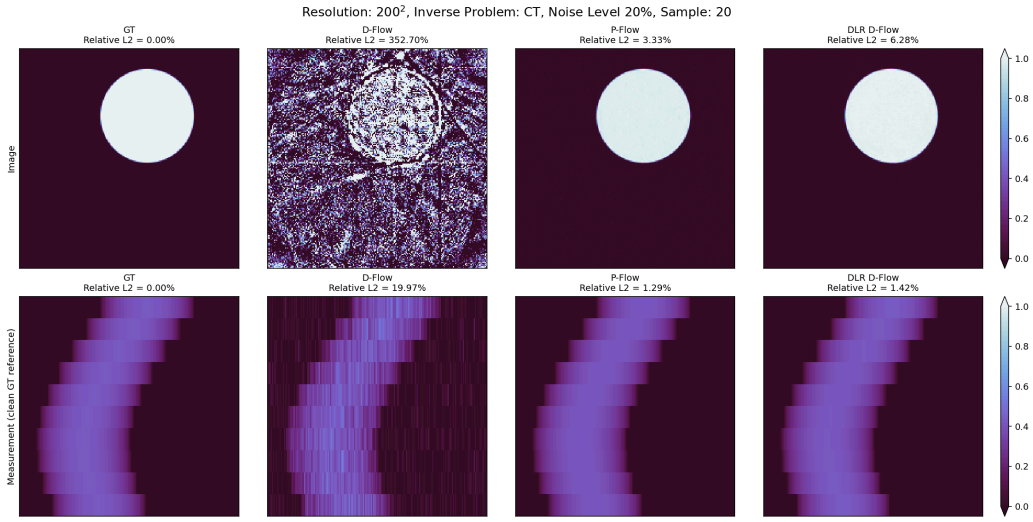


▼ Resolution: 200^2

▼ CT · 23/30 qualifying cases · showing 3 · 20% noise

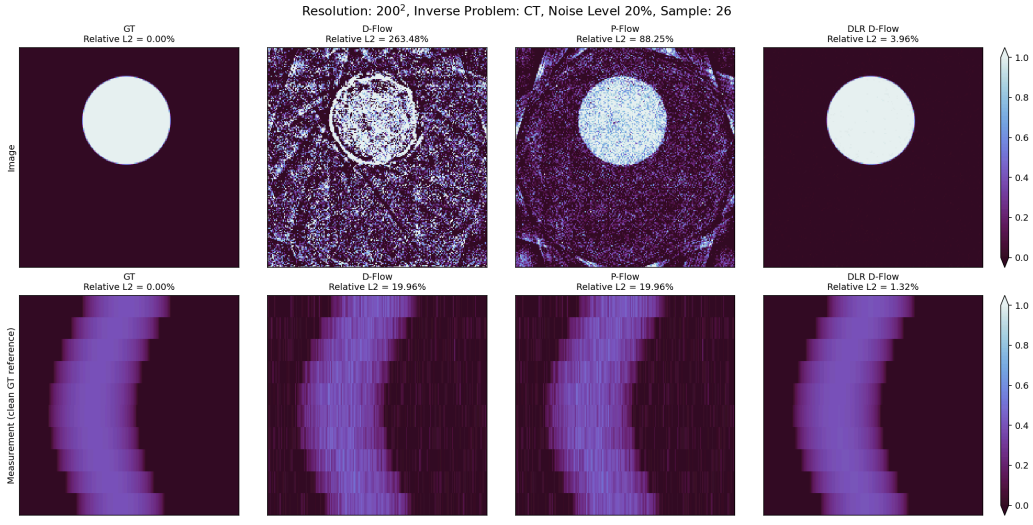
Example 1 · $\Delta = 346.42$ percentage points · CT · 200^2 · Sample 20

D-Flow: **352.70%** · P-Flow: **3.33%** · DLR D-Flow: **6.28%**



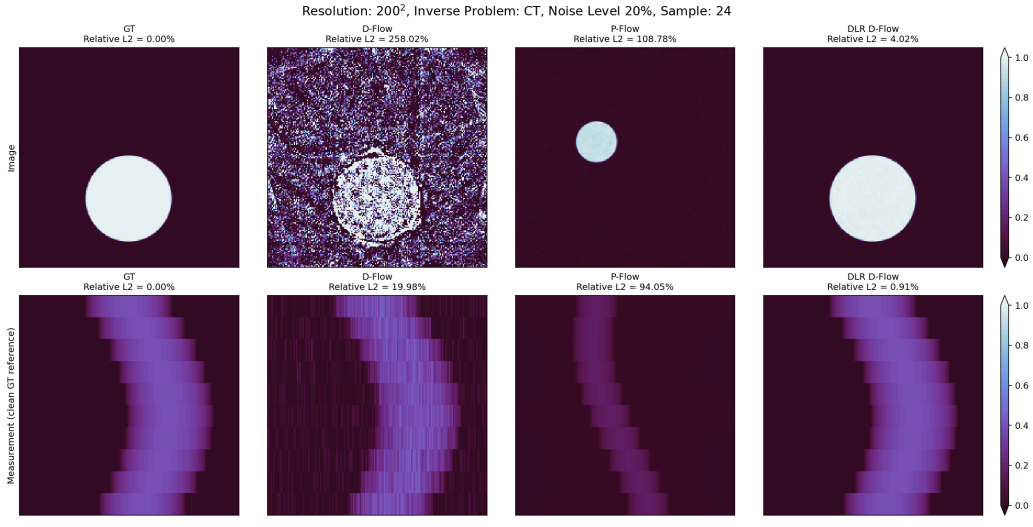
Example 2 · $\Delta = 259.52$ percentage points · CT · 200^2 · Sample 26

D-Flow: **263.48%** · P-Flow: **88.25%** · DLR D-Flow: **3.96%**



Example 3 · $\Delta = 254.00$ percentage points · CT · 200^2 · Sample 24

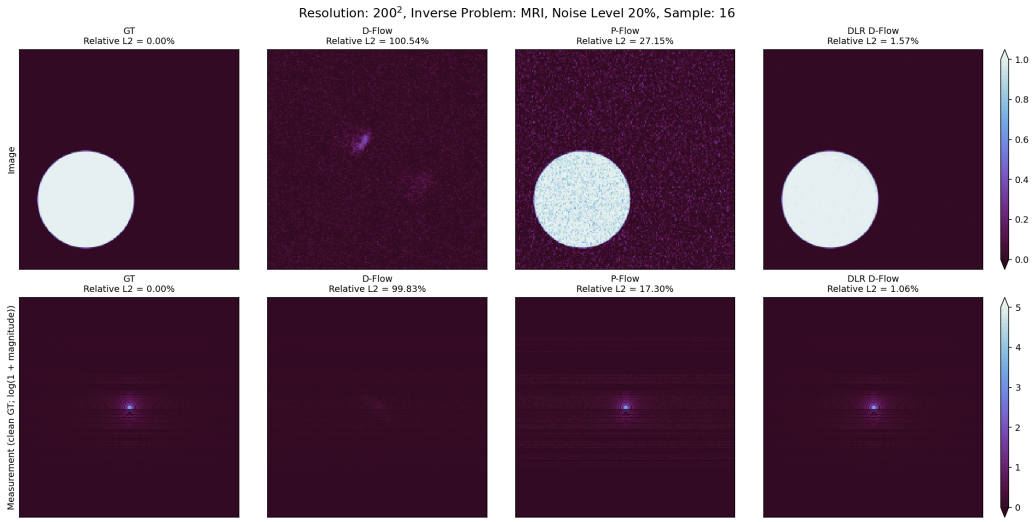
D-Flow: **258.02%** · P-Flow: **108.78%** · DLR D-Flow: **4.02%**



▼ MRI · 21/30 qualifying cases · showing 3 · 20% noise

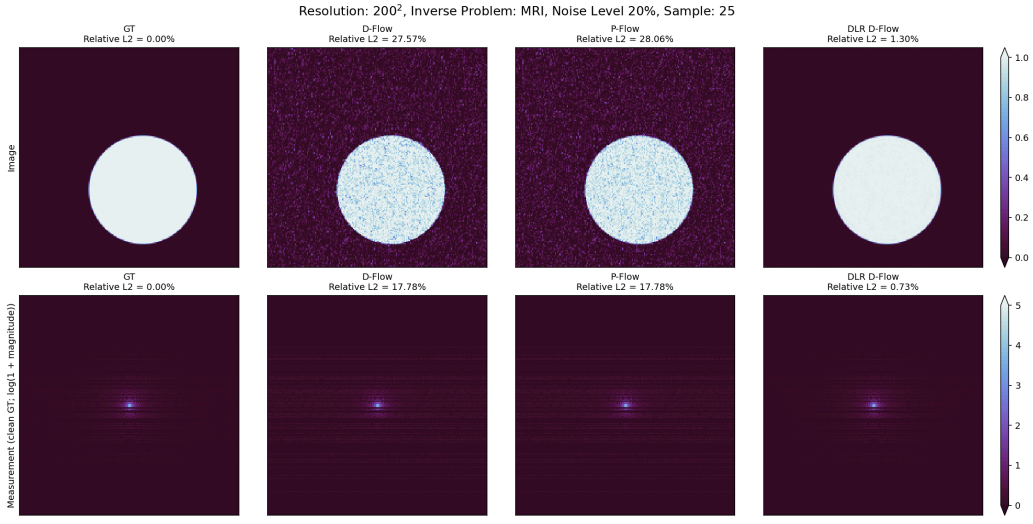
Example 1 · $\Delta = 98.98$ percentage points · MRI · 200^2 · Sample 16

D-Flow: **100.54%** · P-Flow: **27.15%** · DLR D-Flow: **1.57%**



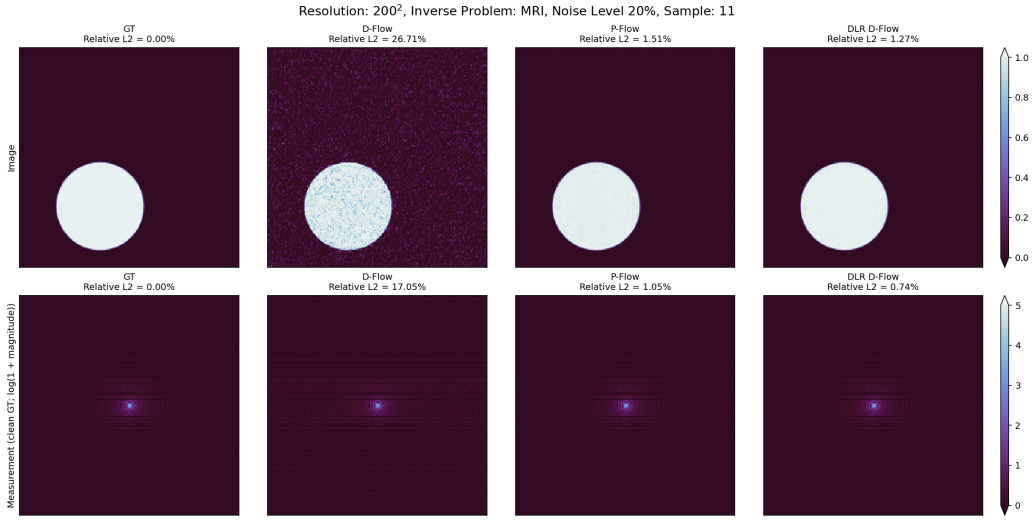
Example 2 · $\Delta = 26.27$ percentage points · MRI · 200^2 · Sample 25

D-Flow: **27.57%** · P-Flow: **28.06%** · DLR D-Flow: **1.30%**



Example 3 · $\Delta = 25.44$ percentage points · MRI · 200^2 · Sample 11

D-Flow: **26.71%** · P-Flow: **1.51%** · DLR D-Flow: **1.27%**

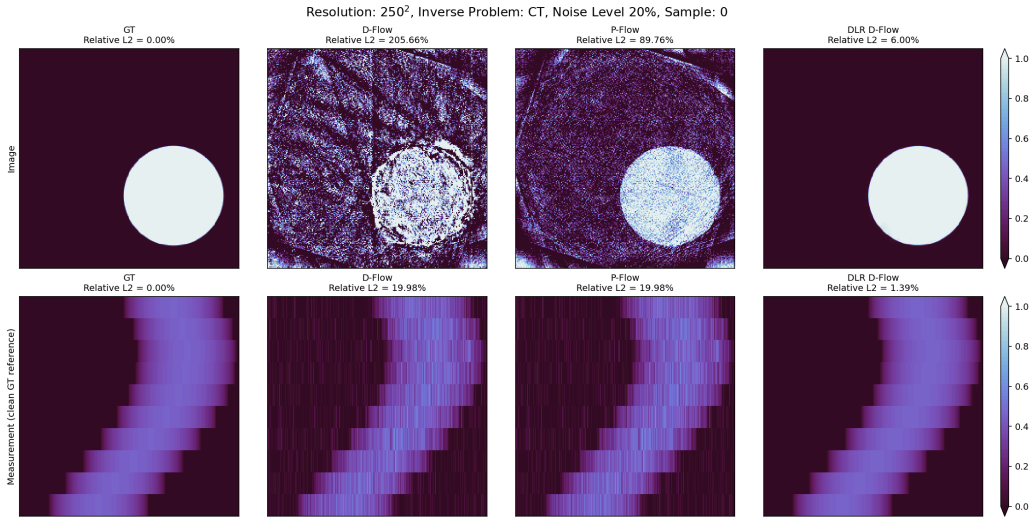


▼ Resolution: 250^2

▼ CT · 21/30 qualifying cases · showing 3 · 20% noise

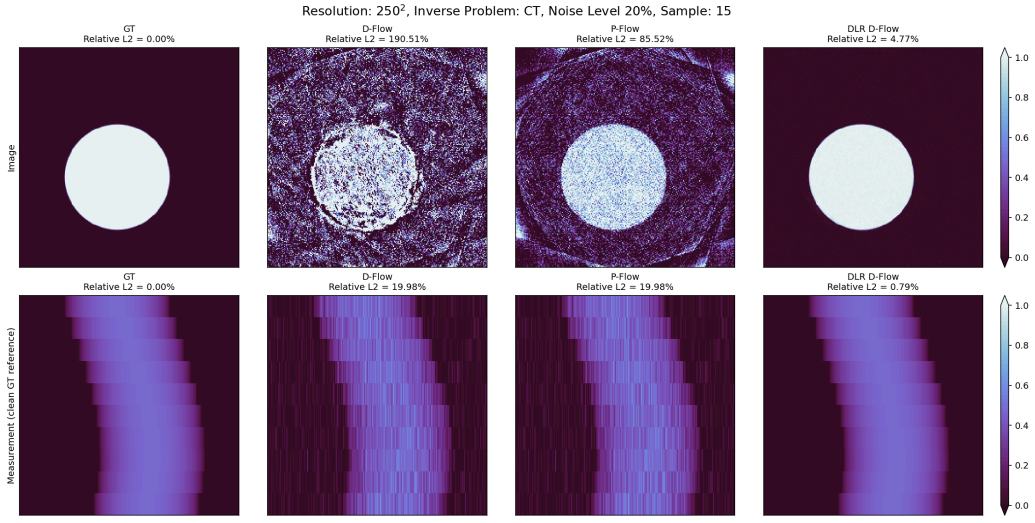
Example 1 · $\Delta = 199.66$ percentage points · CT · 250^2 · Sample 0

D-Flow: **205.66%** · P-Flow: **89.76%** · DLR D-Flow: **6.00%**



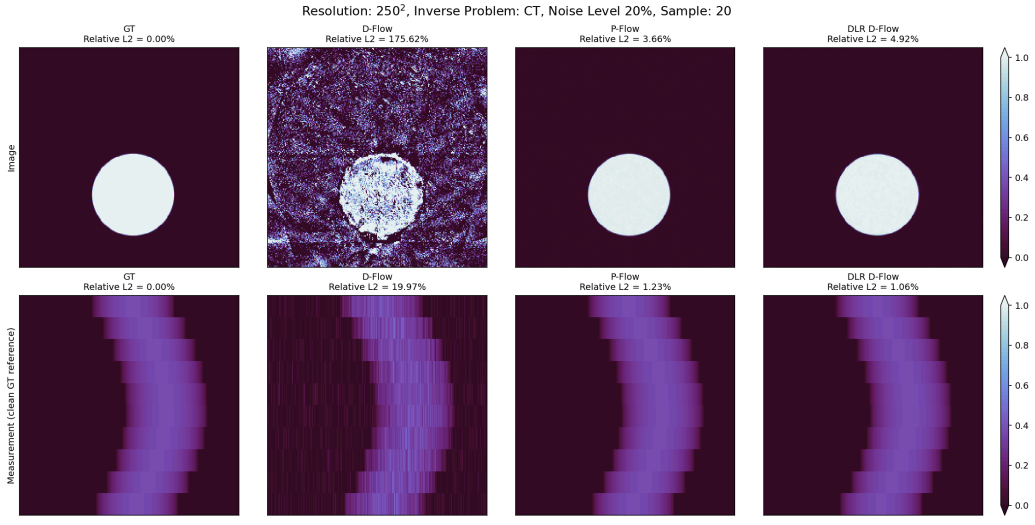
Example 2 · $\Delta = 185.74$ percentage points · CT · 250^2 · Sample 15

D-Flow: **190.51%** · P-Flow: **85.52%** · DLR D-Flow: **4.77%**



Example 3 · $\Delta = 170.69$ percentage points · CT · 250^2 · Sample 20

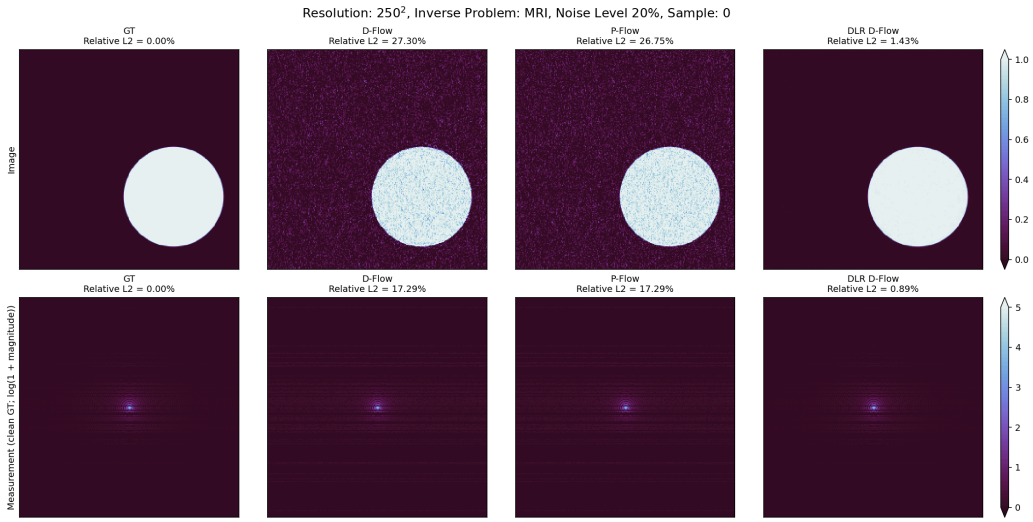
D-Flow: **175.62%** · P-Flow: **3.66%** · DLR D-Flow: **4.92%**



▼ MRI · 21/30 qualifying cases · showing 3 · 20% noise

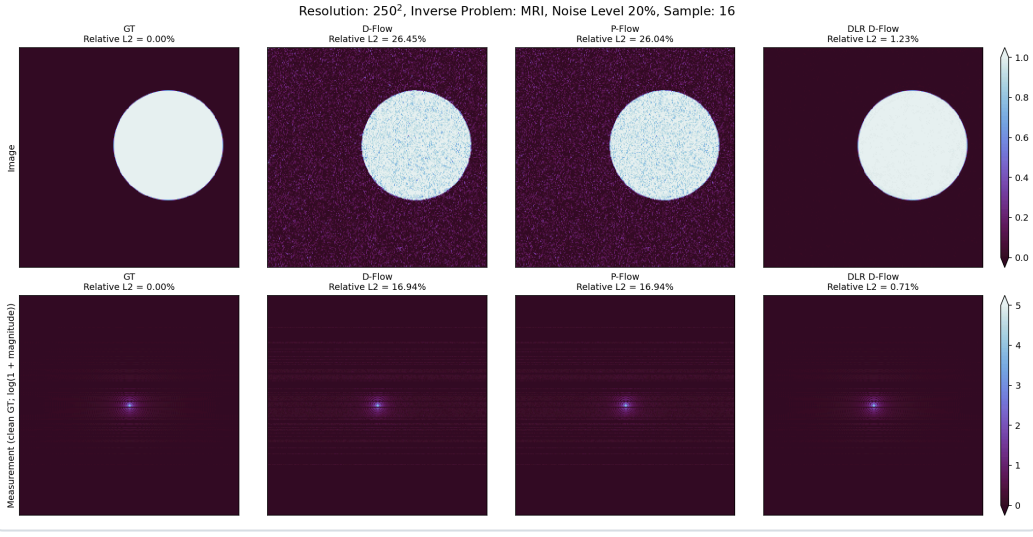
Example 1 · $\Delta = 25.87$ percentage points · MRI · 250^2 · Sample 0

D-Flow: **27.30%** · P-Flow: **26.75%** · DLR D-Flow: **1.43%**



Example 2 · $\Delta = 25.22$ percentage points · MRI · 250^2 · Sample 16

D-Flow: **26.45%** · P-Flow: **26.04%** · DLR D-Flow: **1.23%**



Example 3 · $\Delta = 25.15$ percentage points · MRI · 250^2 · Sample 6

D-Flow: **26.27%** · P-Flow: **25.91%** · DLR D-Flow: **1.11%**

